



BobSim

FourPostEval Suspension Kinematics and Compliance

Heave and roll sweeps

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Run Notes

- Heave and roll sweeps evaluate camber, toe, caster, KPI, trail, scrub, motion ratio, and jacking characteristics.
- Summary metrics use the same KnC-style table layout as the legacy report.

FourPostEval Setup Summary

Sprung mass 226.4 kg | CG (-0.810, 0.000, 0.292) m | Wheelbase 1.549 m | Track Fr/Rr 1.212 / 1.212 m
CG bias Fr/Rr 47.7 / 52.3% | CG bias L/R 50.0 / 50.0%

Front Setup				Rear Setup			
FL / FR				RL / RR			
Metric	FL	FR	Units	Metric	RL	RR	Units
Sprung mass	54.0	54.0	kg	Sprung mass	59.2	59.2	kg
Unsprung mass	9.0	9.0	kg	Unsprung mass	8.3	8.3	kg
Sprung load	530	530	N	Sprung load	580	580	N
Motion ratio	1.004	1.004	wheel/spring	Motion ratio	1.238	1.238	wheel/spring
Spring rate	26269	26269	N/m	Spring rate	43782	43782	N/m
Spring force	532	532	N	Spring force	719	719	N
Installed length	179.4	179.4	mm	Installed length	247.0	247.0	mm
Compression	20.2	20.2	mm	Compression	16.4	16.4	mm
Free length	199.7	199.7	mm	Free length	263.4	263.4	mm
Wheel rate	26079	26079	N/m	Wheel rate	28551	28551	N/m
Sprung freq	3.10	3.10	Hz	Sprung freq	3.07	3.07	Hz
Unsprung freq	18.79	18.79	Hz	Unsprung freq	19.80	19.80	Hz

FourPostEval Metrics Summary

Heave Gains

Camber	0.0264	deg/mm
Toe	-0.0002	deg/mm
Caster	-0.0023	deg/mm
KPI	0.0264	deg/mm
Trail	-0.0011	mm/mm
Scrub	0.0018	mm/mm

Anti Metrics

Anti-Dive	3.0	%
Anti-Squat	-4.6	%
Front Anti-Roll	0.1	%
Rear Anti-Roll	0.1	%
LLTD (Front)	52.1	%

Roll Gains

Camber	0.7245	deg/deg
Toe	0.0013	deg/deg
Caster	0.0344	deg/deg
KPI	0.7249	deg/deg
Trail	-0.0262	mm/deg
Scrub	0.0455	mm/deg

Motion Ratios

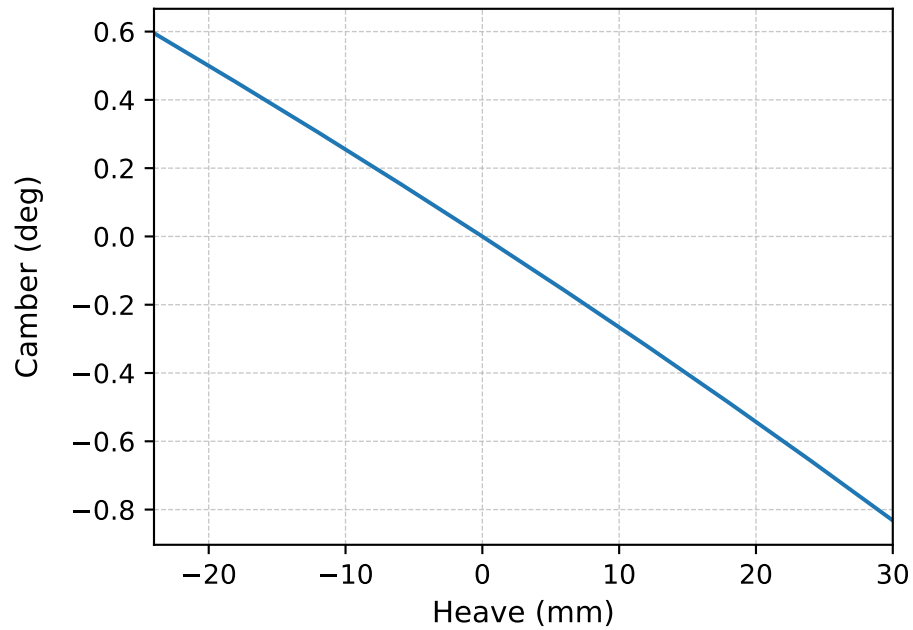
Front MR	1.002	wheel/spring
Rear MR	1.255	wheel/spring
Front Bar MR	0.127	roll/stabar
Rear Bar MR	0.155	roll/stabar

Roll Stiffness

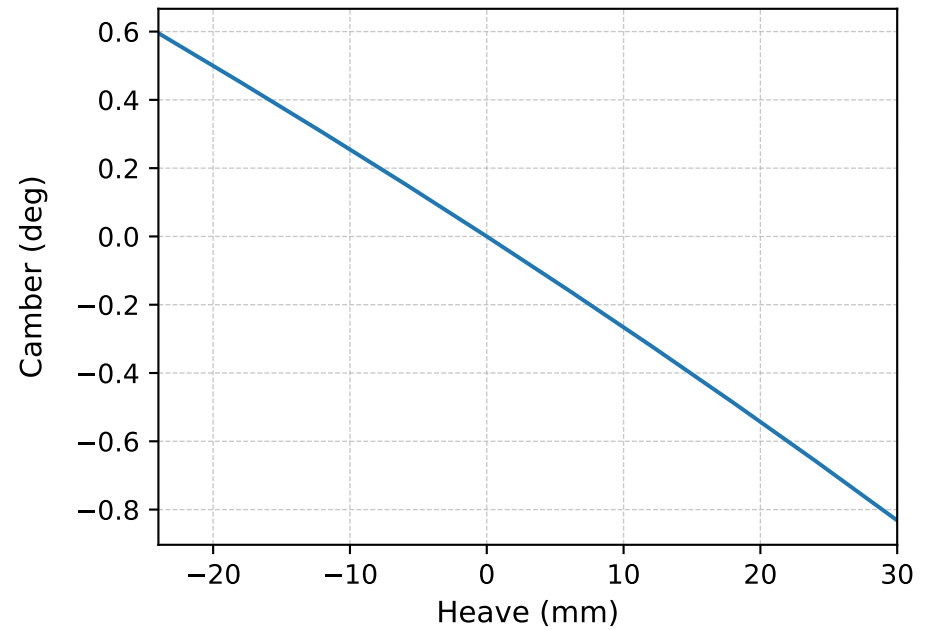
Spring Front	376	Nm/deg
Spring Rear	224	Nm/deg
ARB Front	809	Nm/deg
ARB Rear	868	Nm/deg
Total Front	1184	Nm/deg
Total Rear	1092	Nm/deg

Camber vs Heave

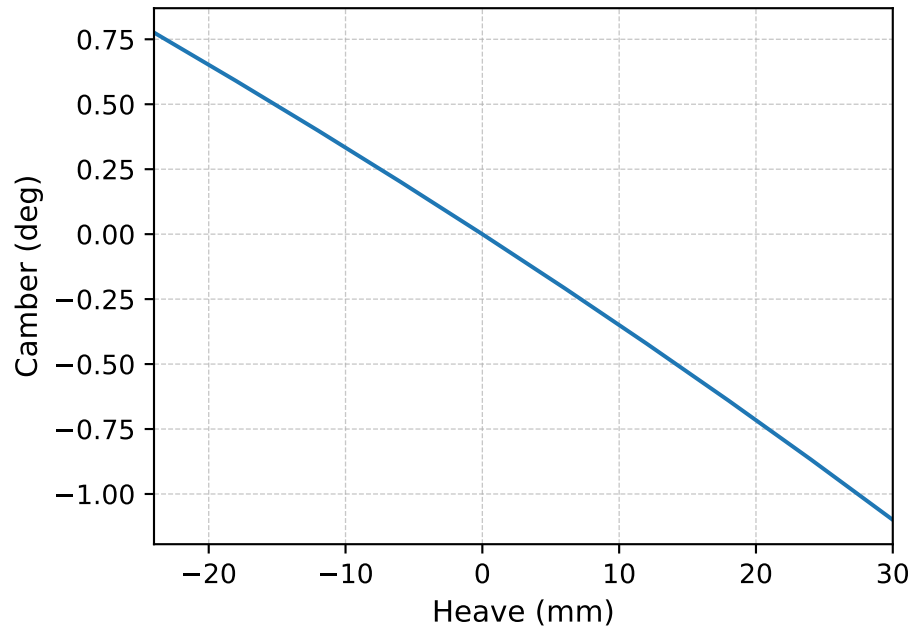
Front Left



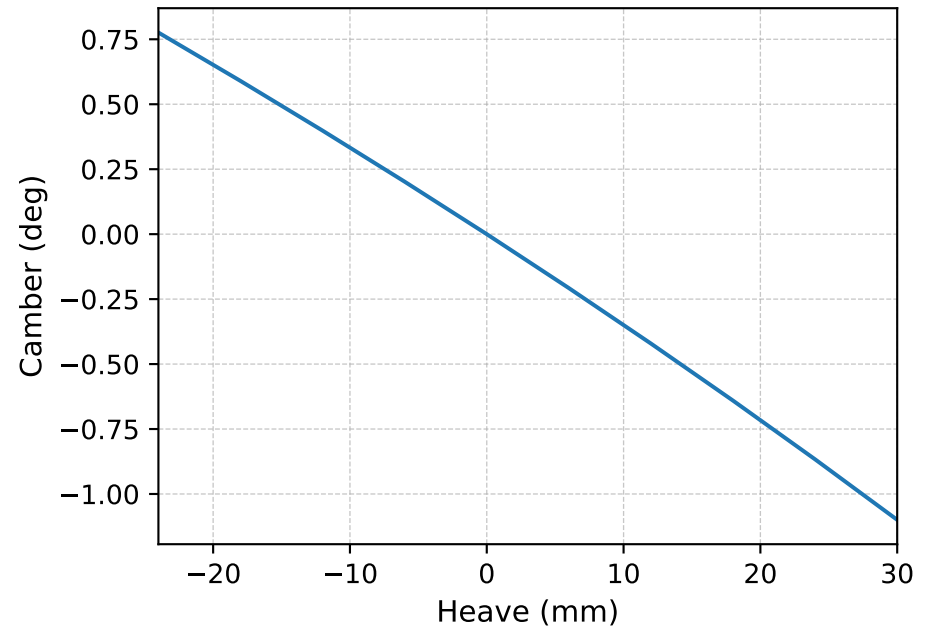
Front Right



Rear Left

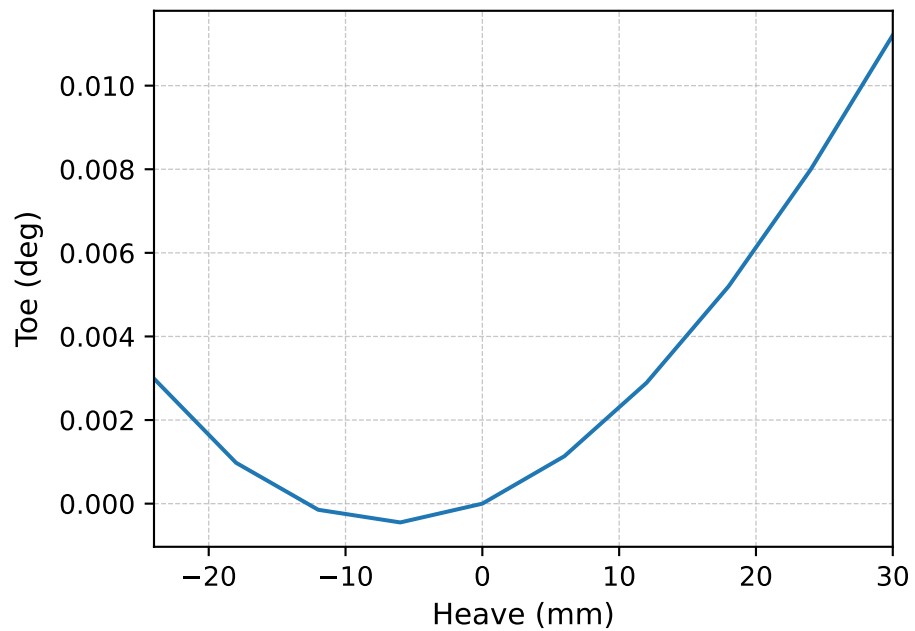


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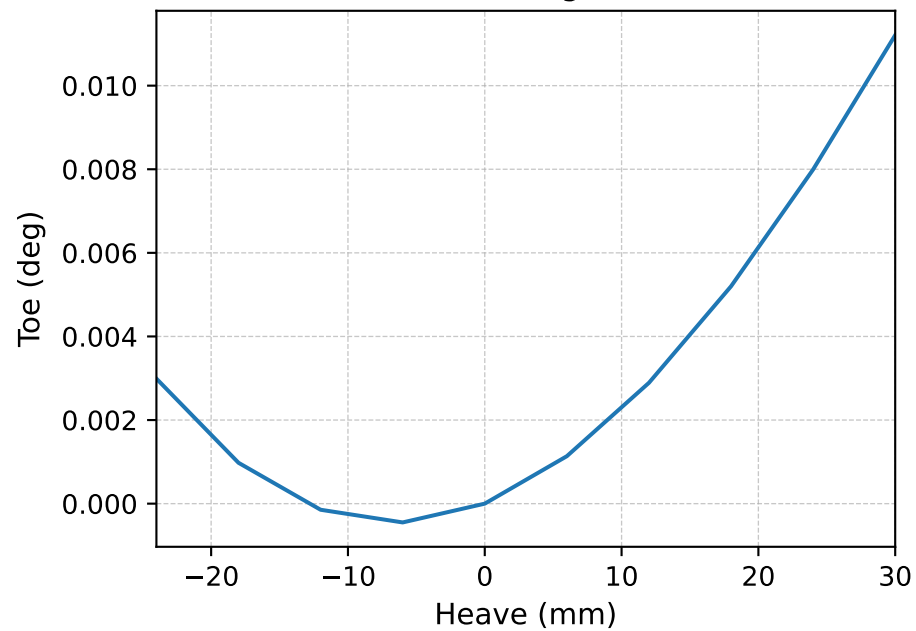


Toe vs Heave

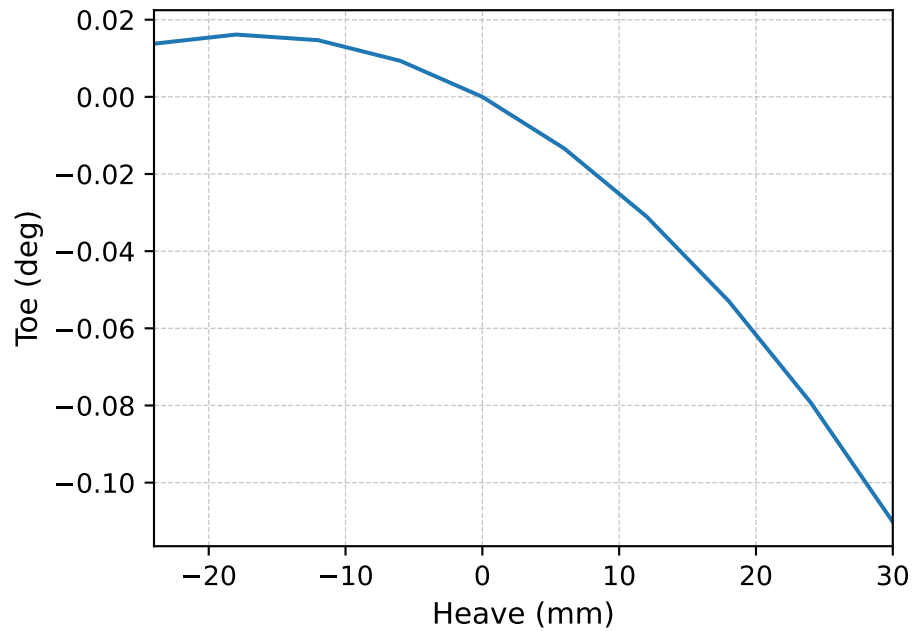
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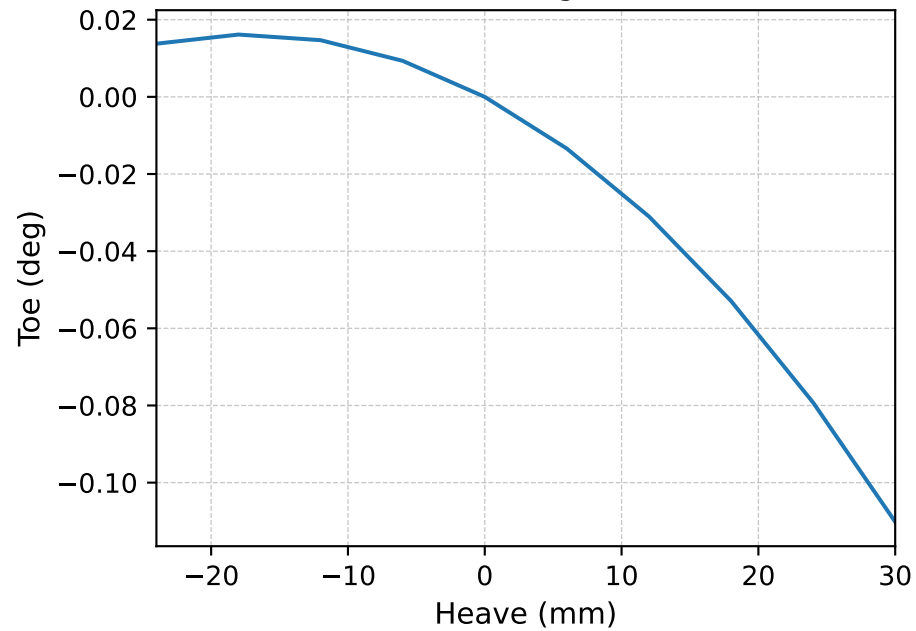
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Rear Left

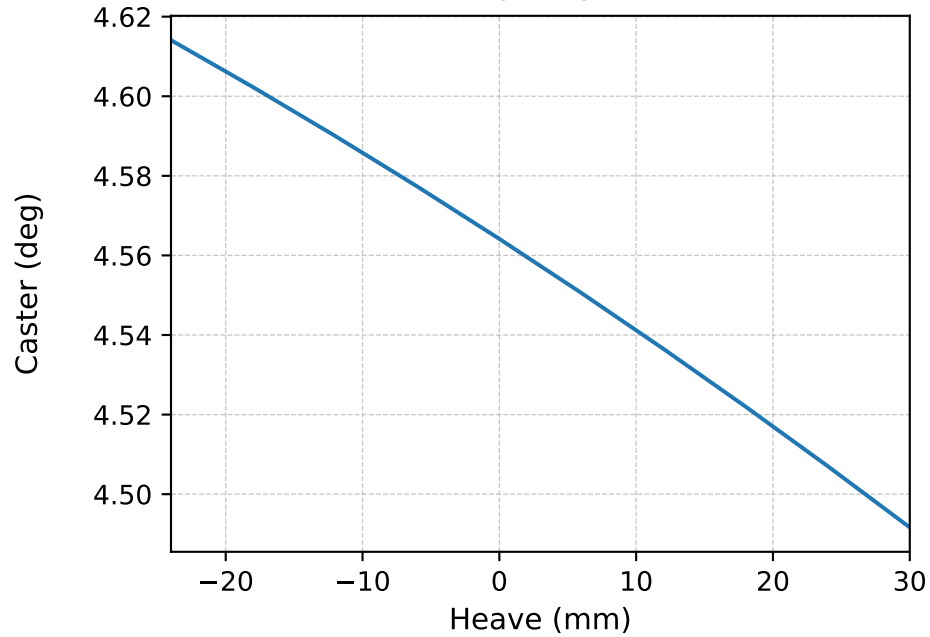


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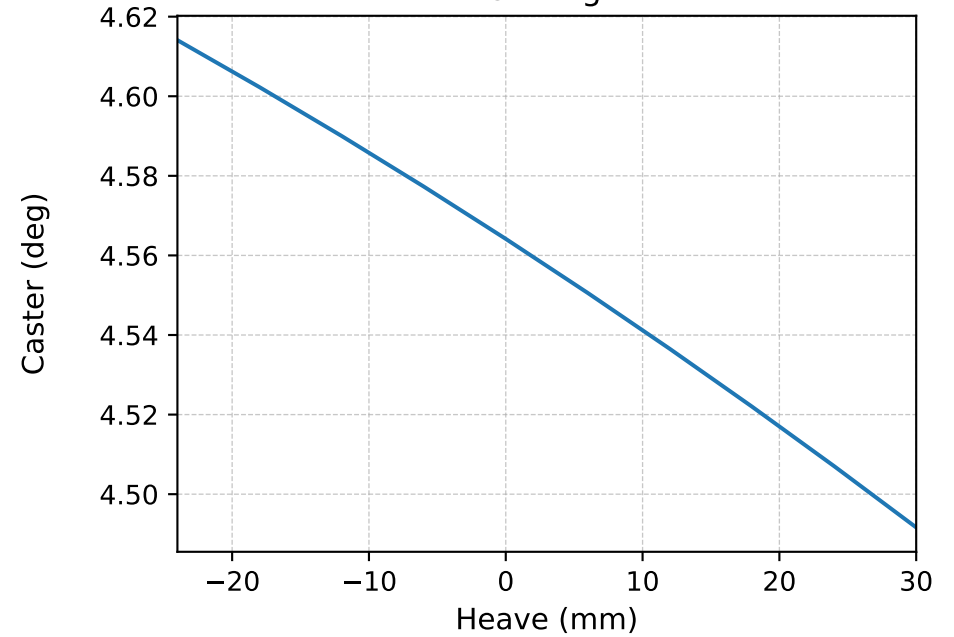


Caster vs Heave

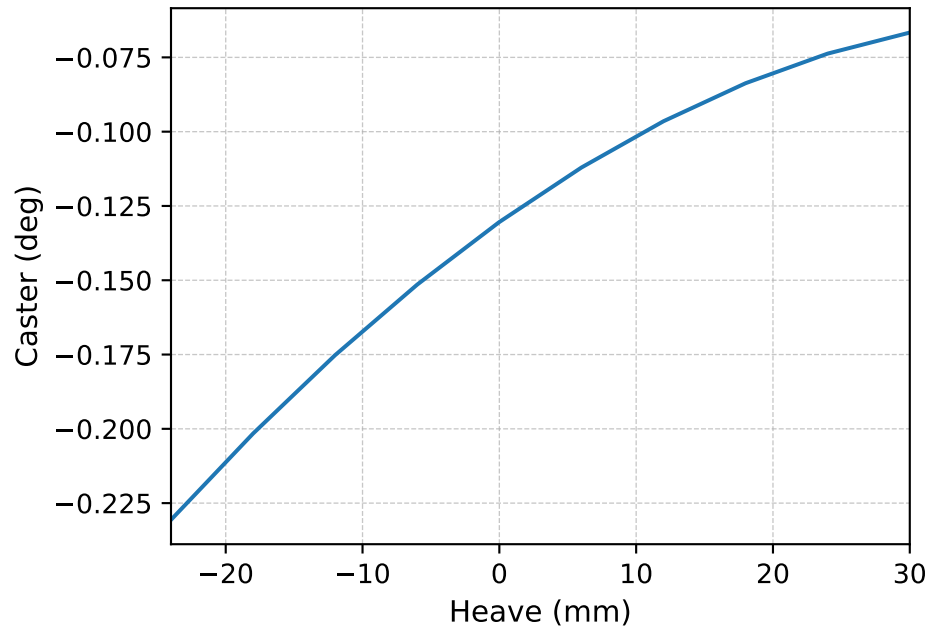
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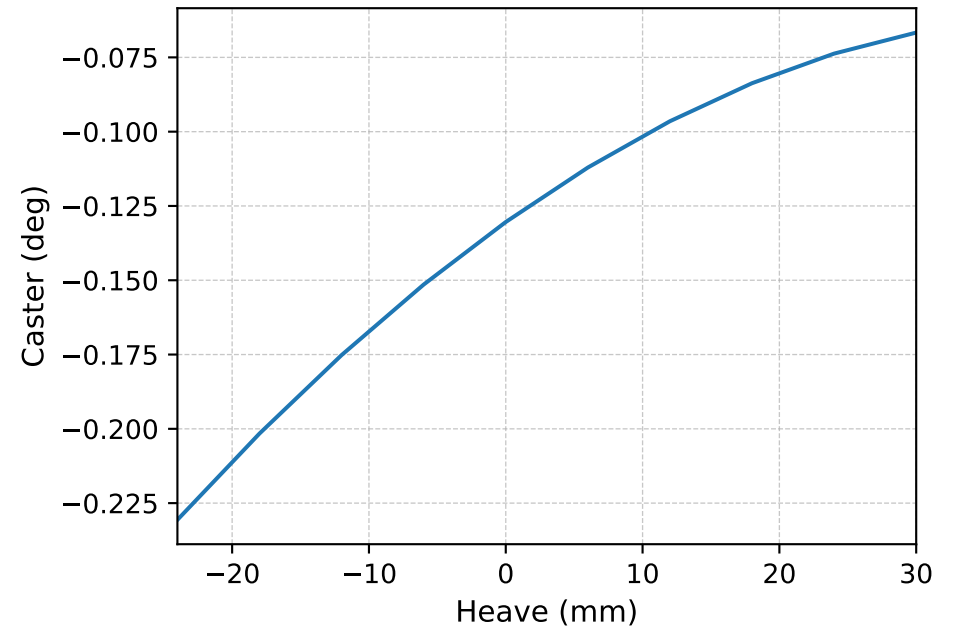
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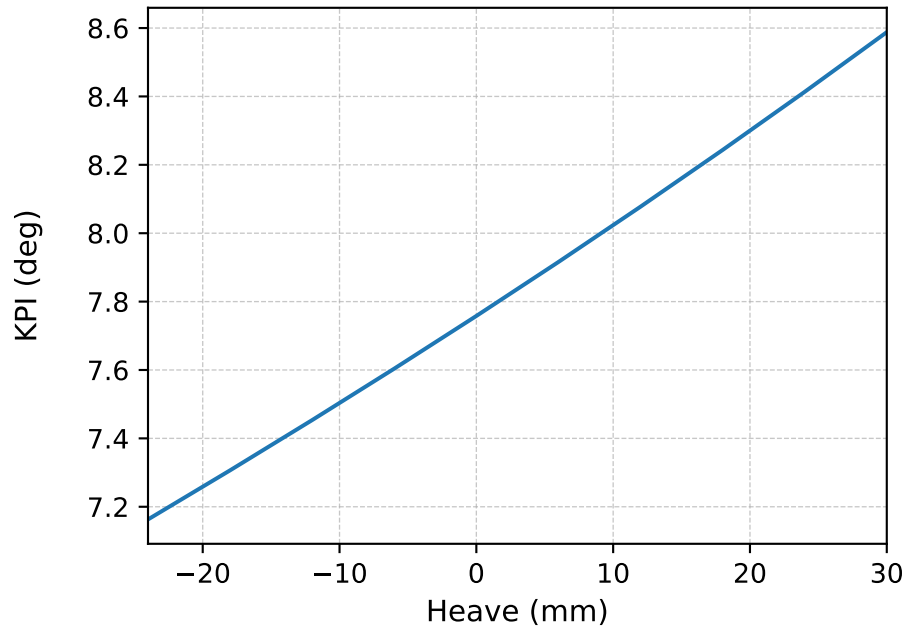


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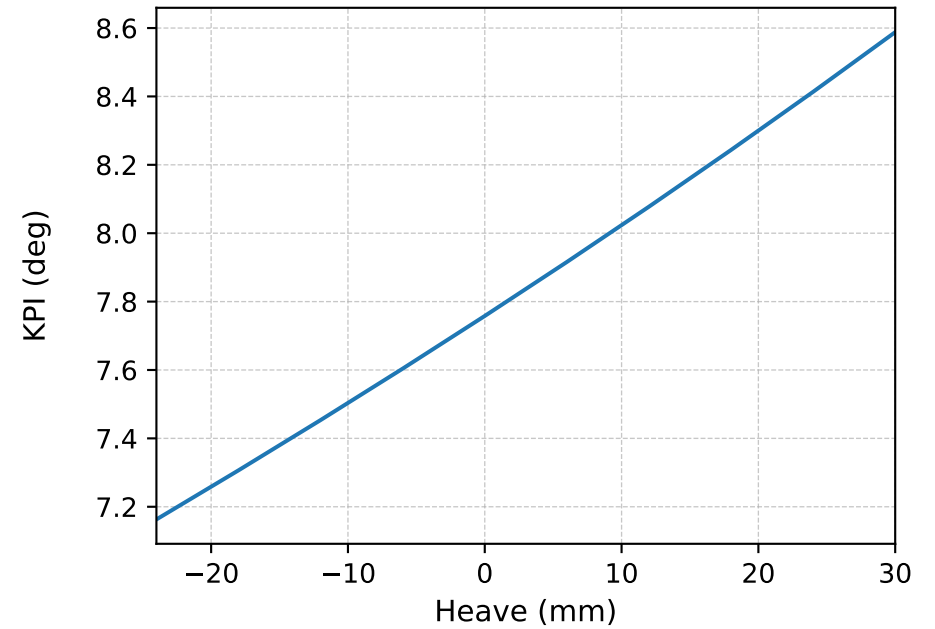


KPI vs Heave

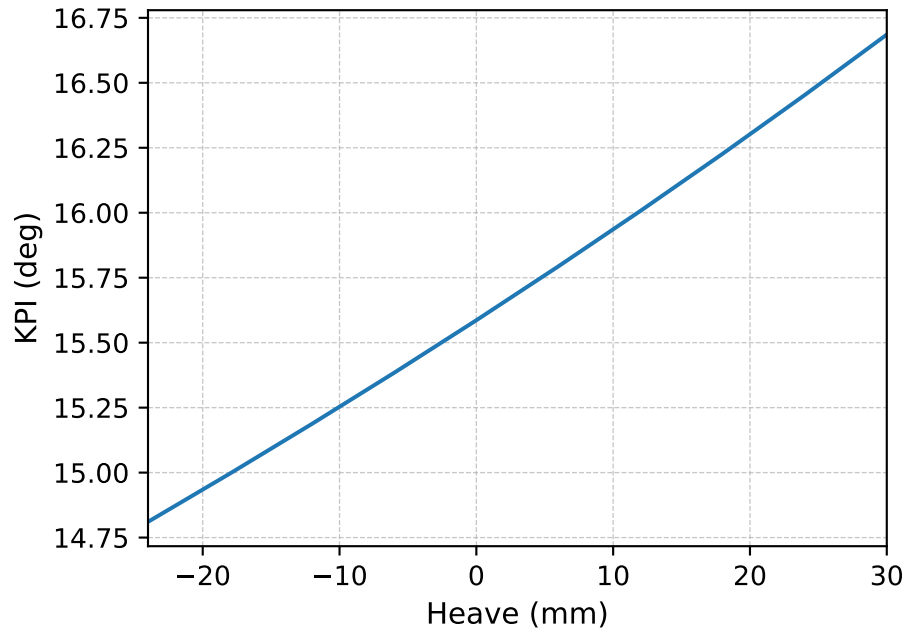
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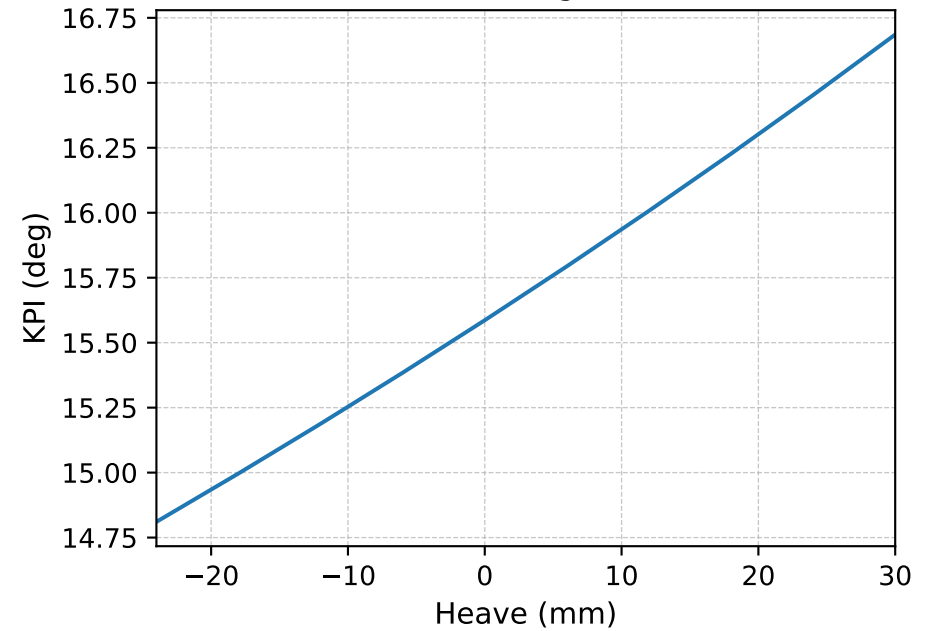
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Rear Left

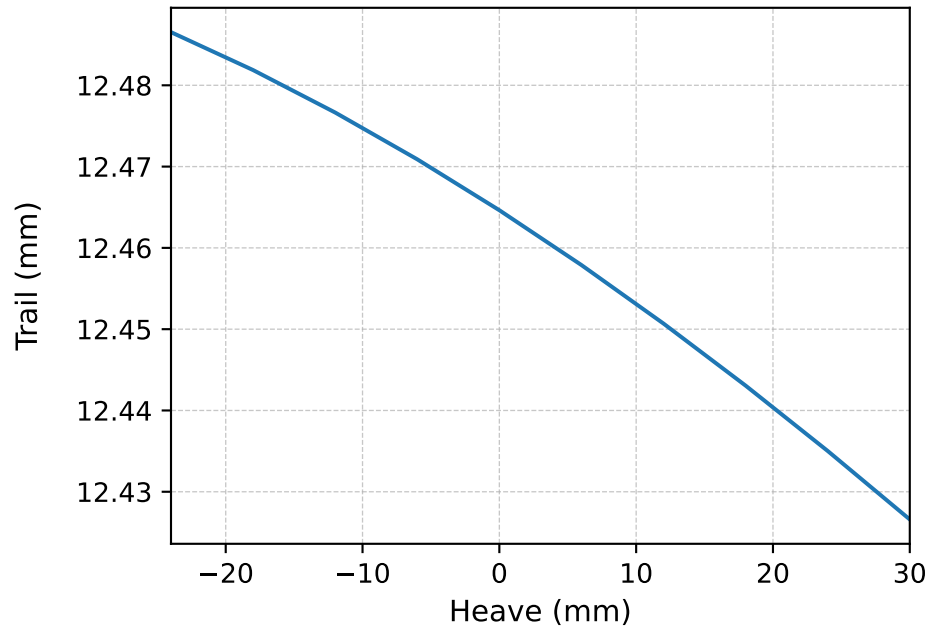


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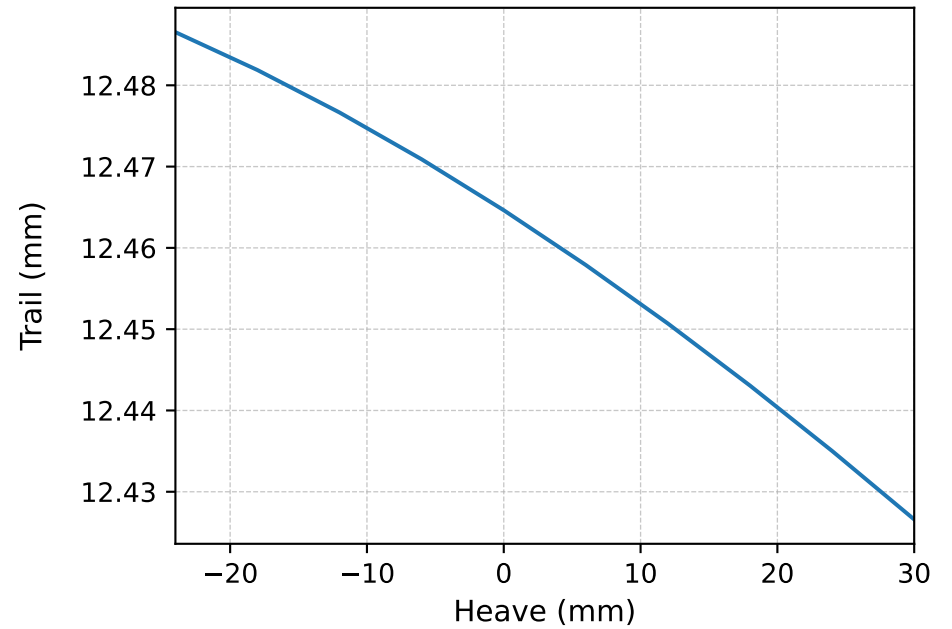


Mechanical Trail vs Heave

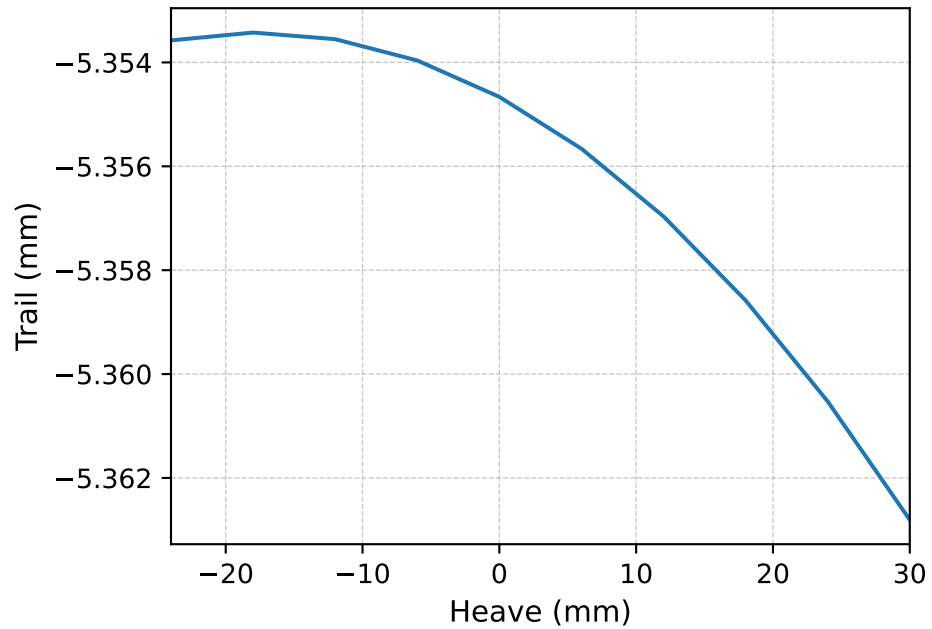
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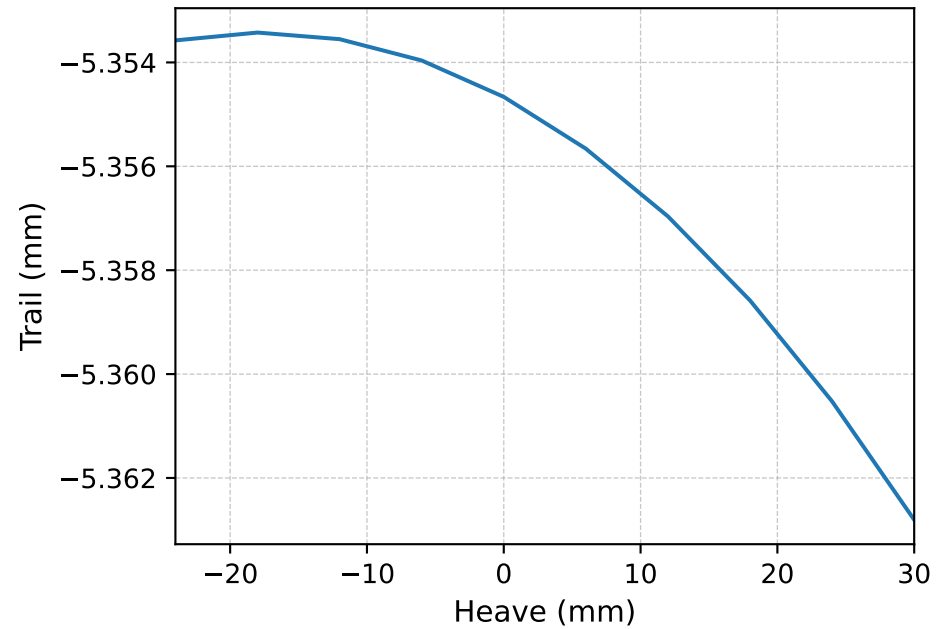
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Rear Left

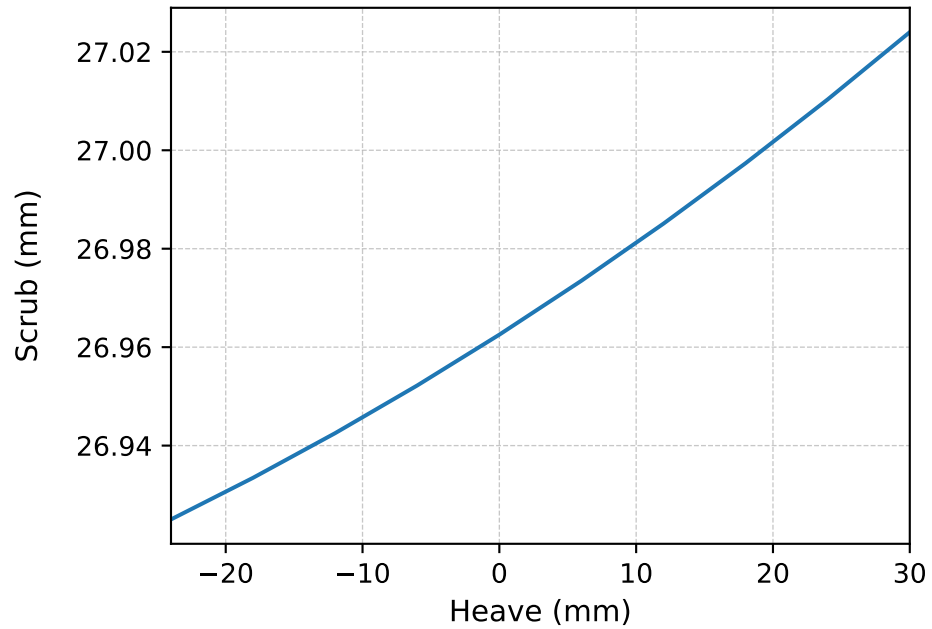


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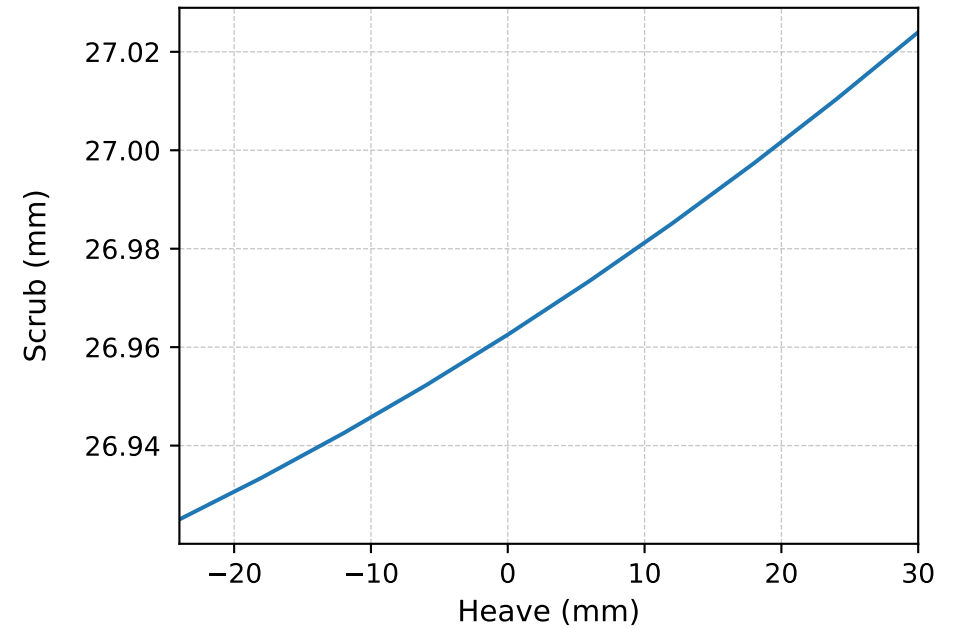


Mechanical Scrub vs Heave

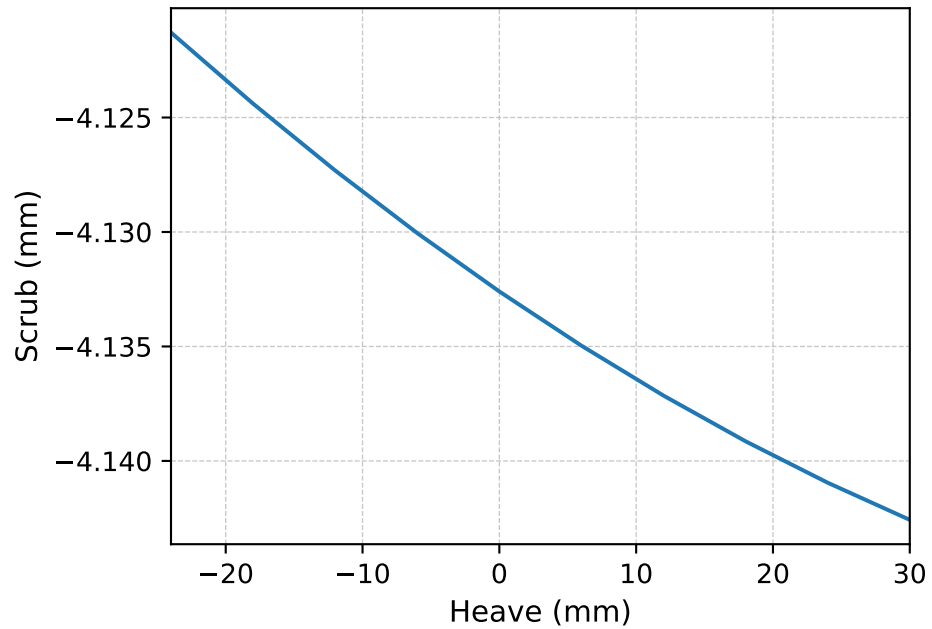
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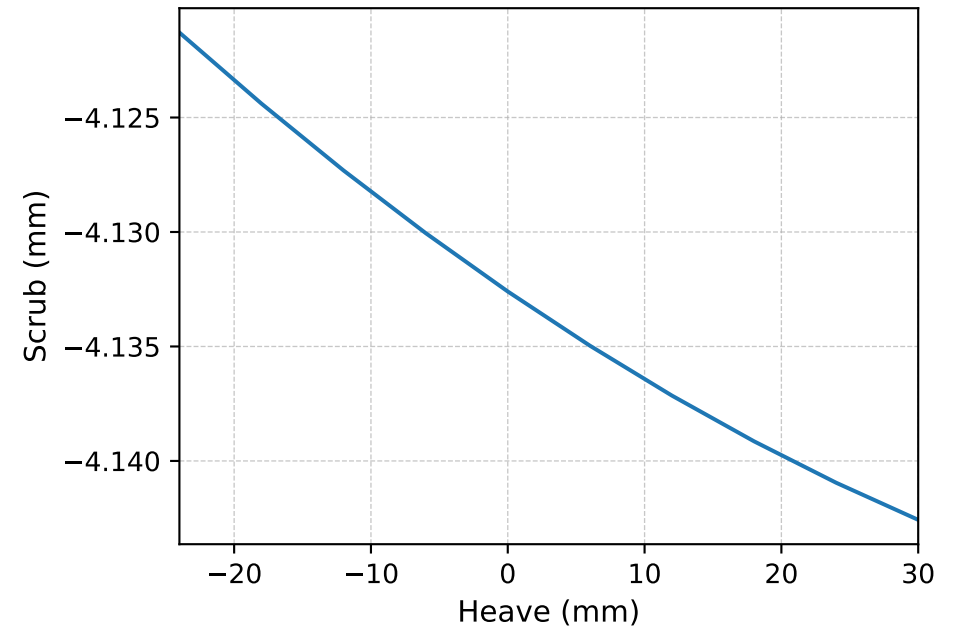
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Rear Left

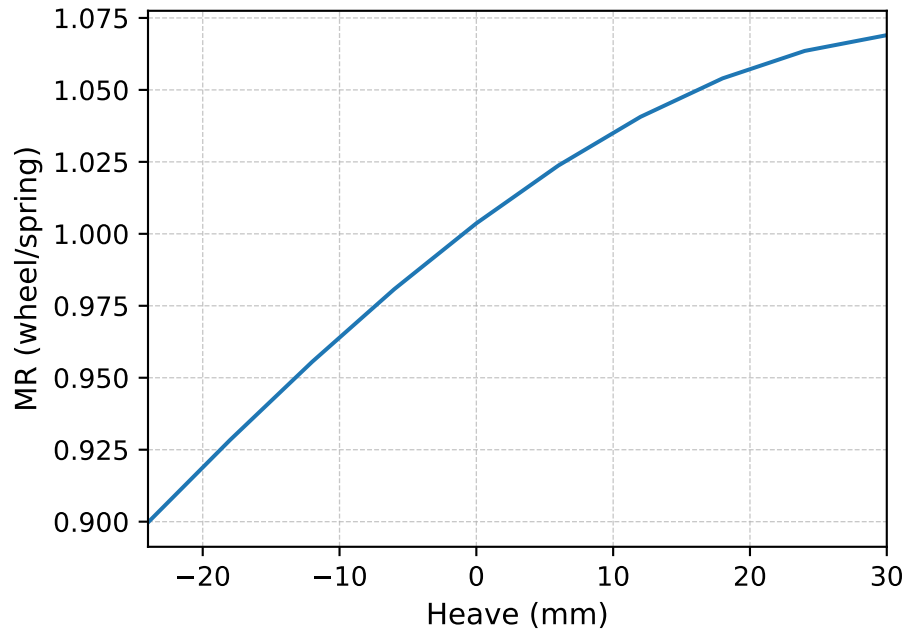


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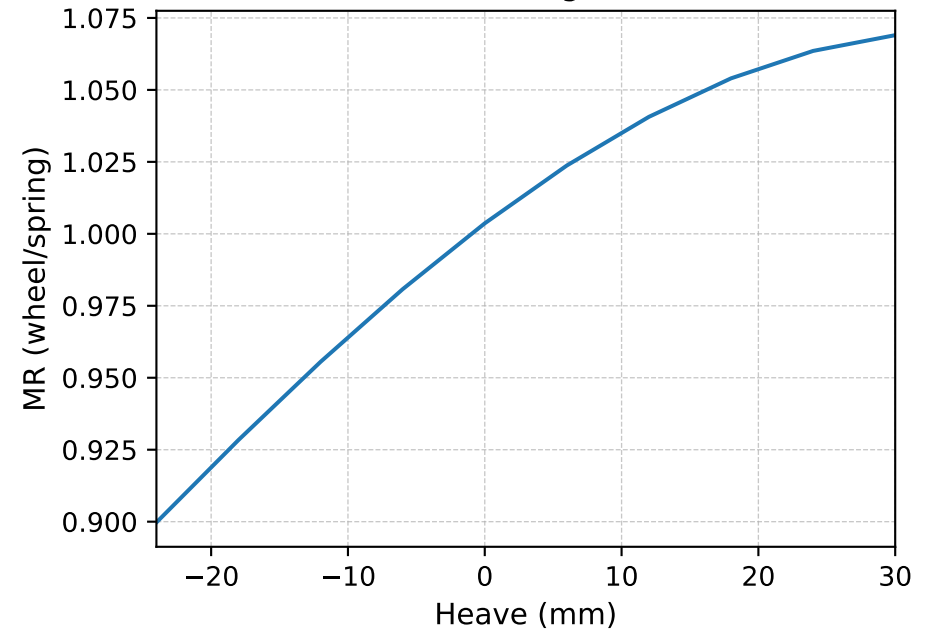


Motion Ratio vs Heave

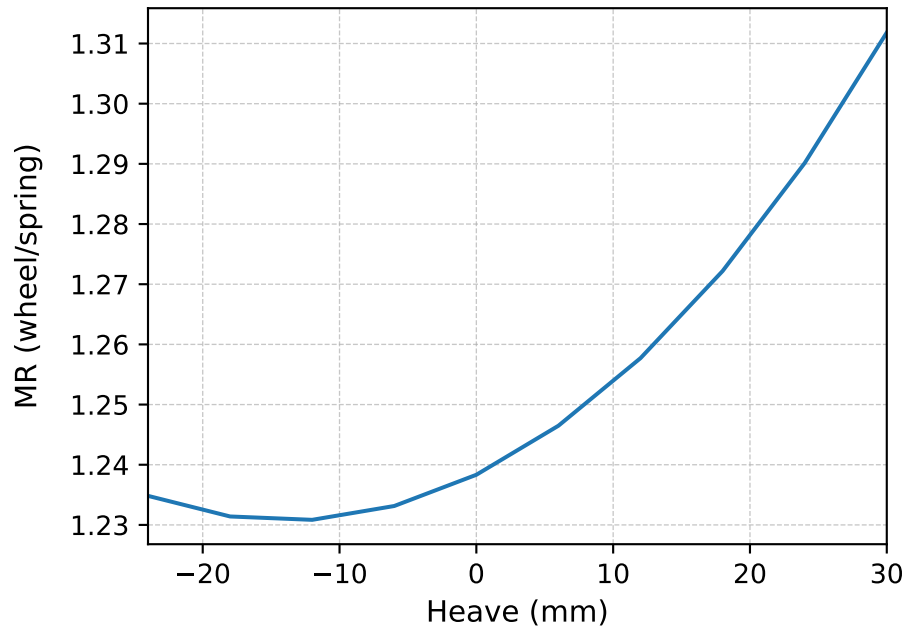
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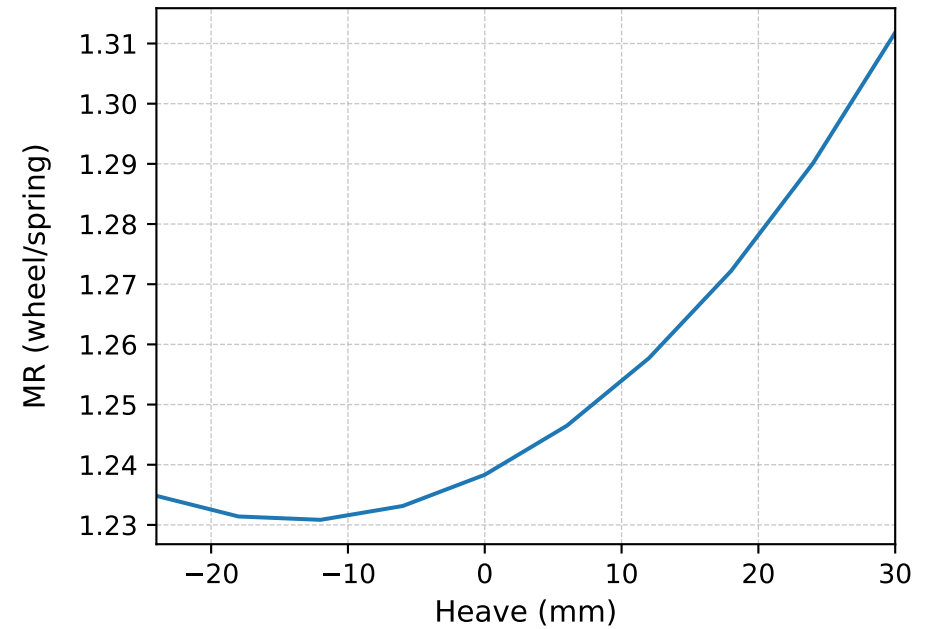
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Rear Left

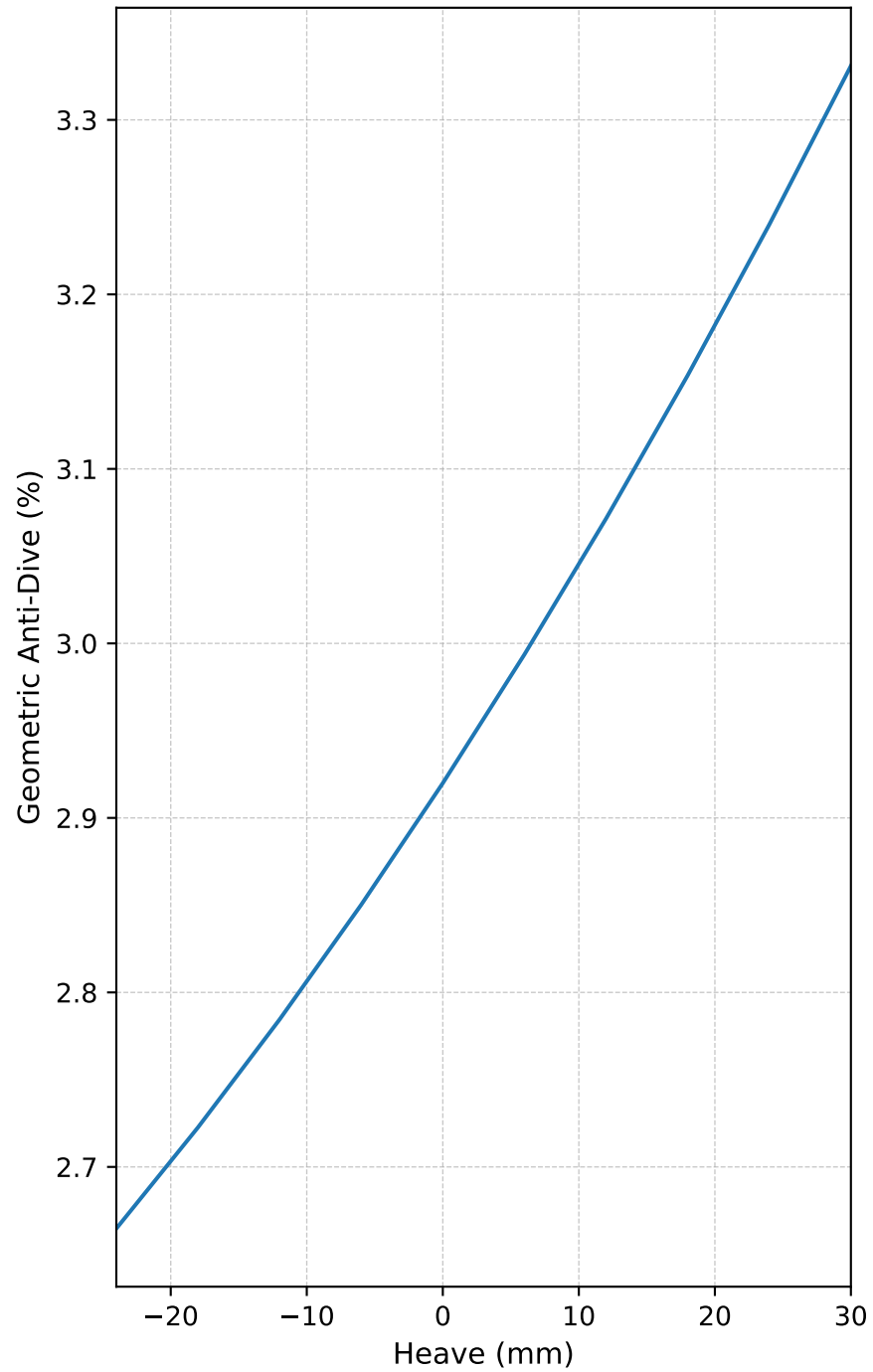


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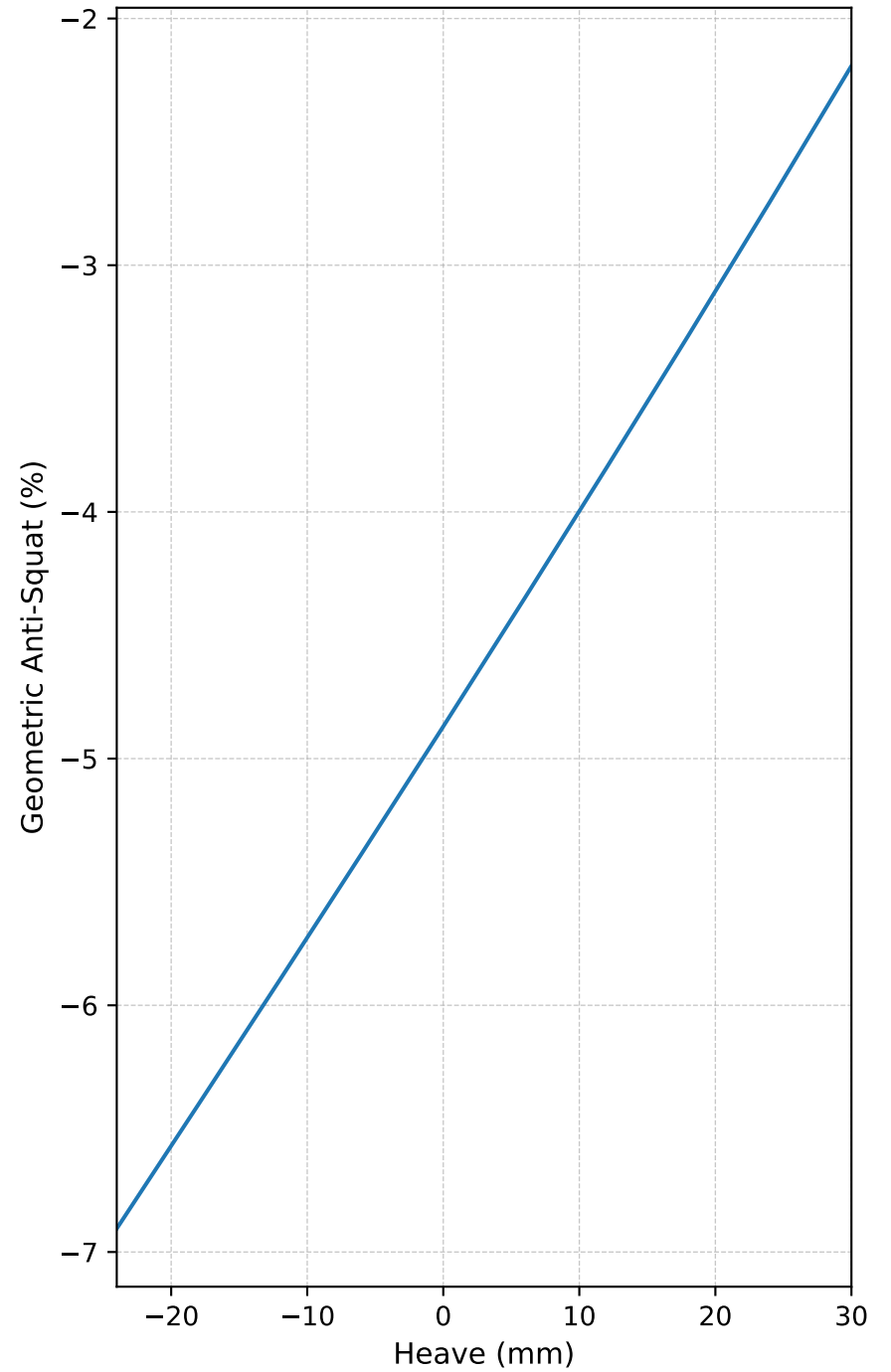


Anti vs Heave

Front

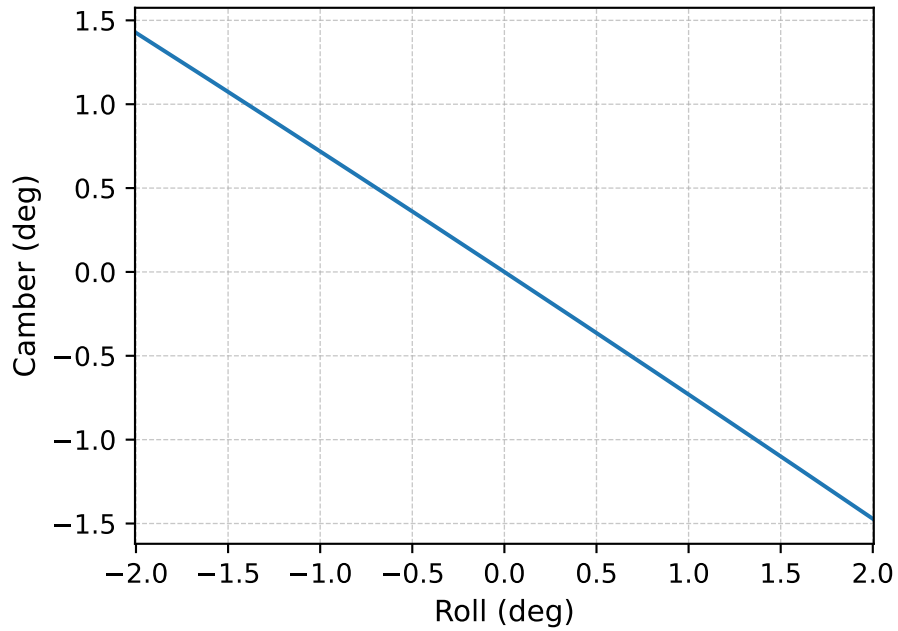


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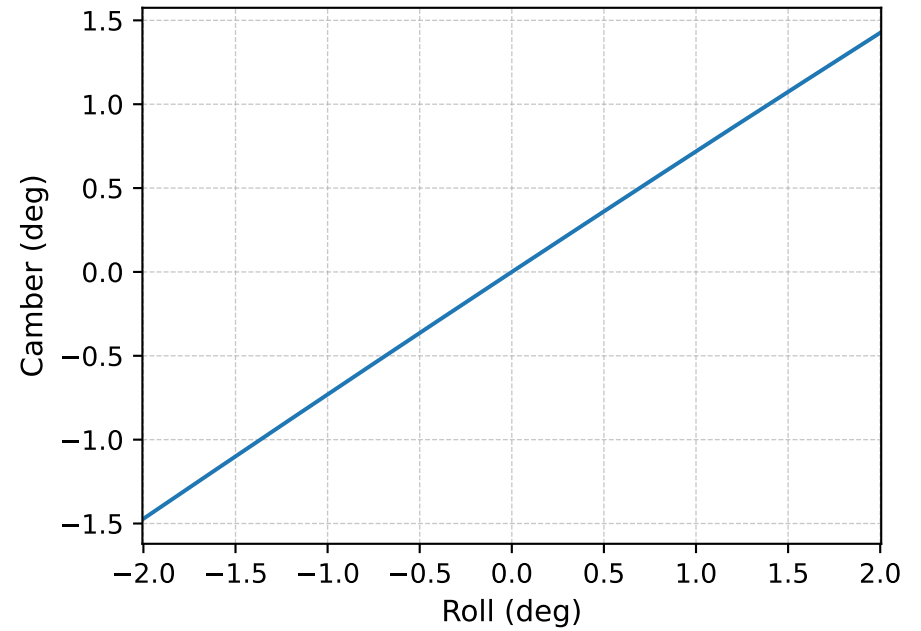


Camber vs Roll

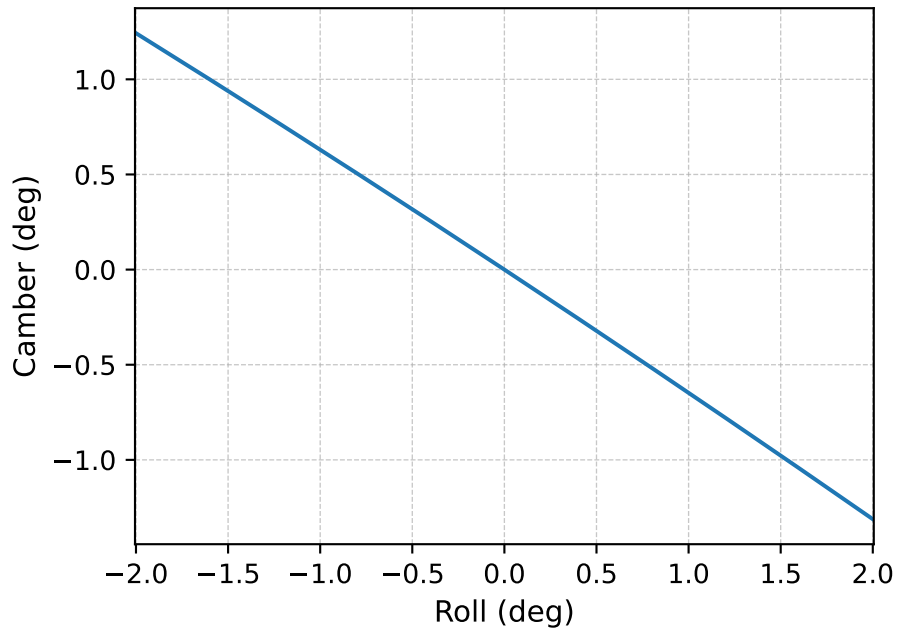
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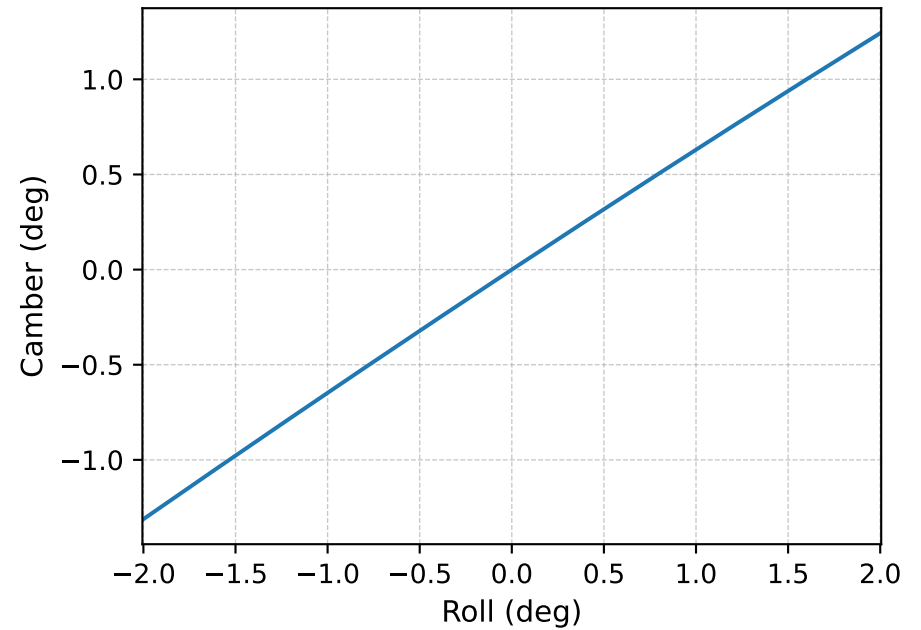
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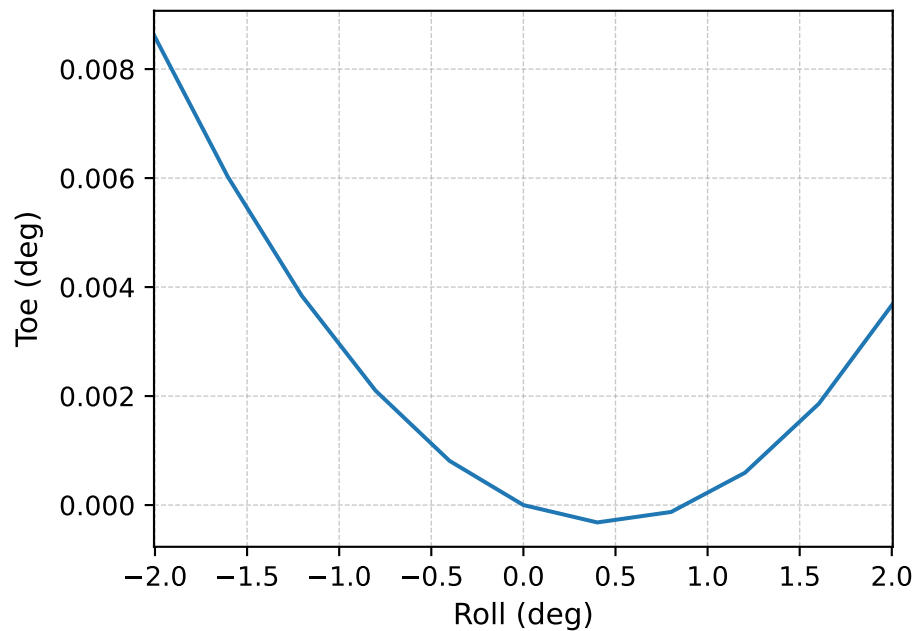


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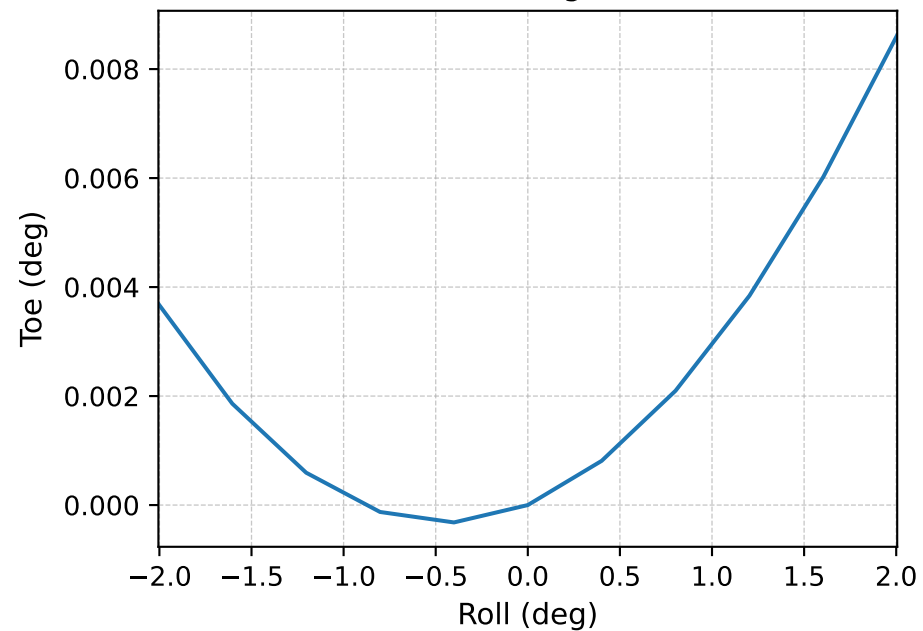


Toe vs Roll

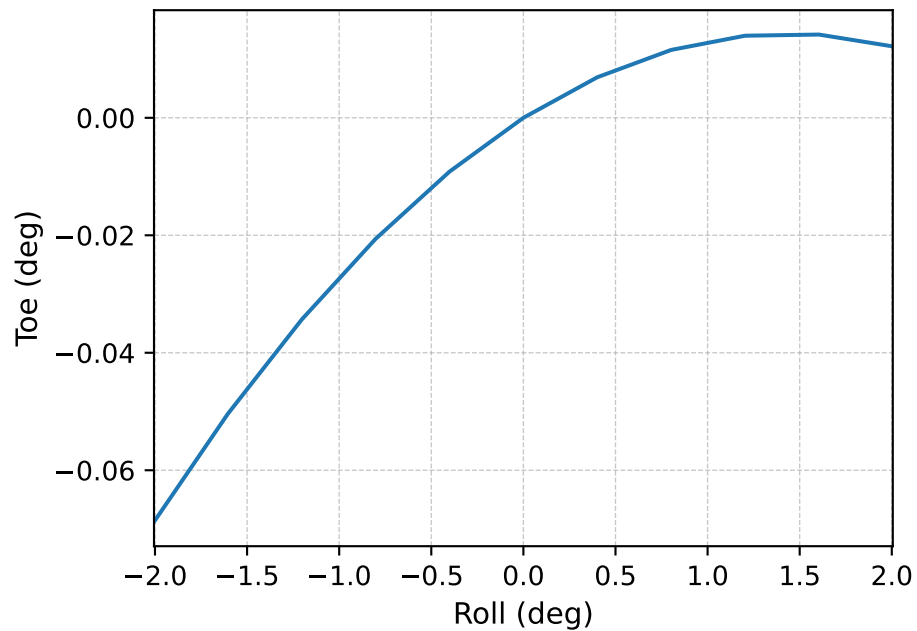
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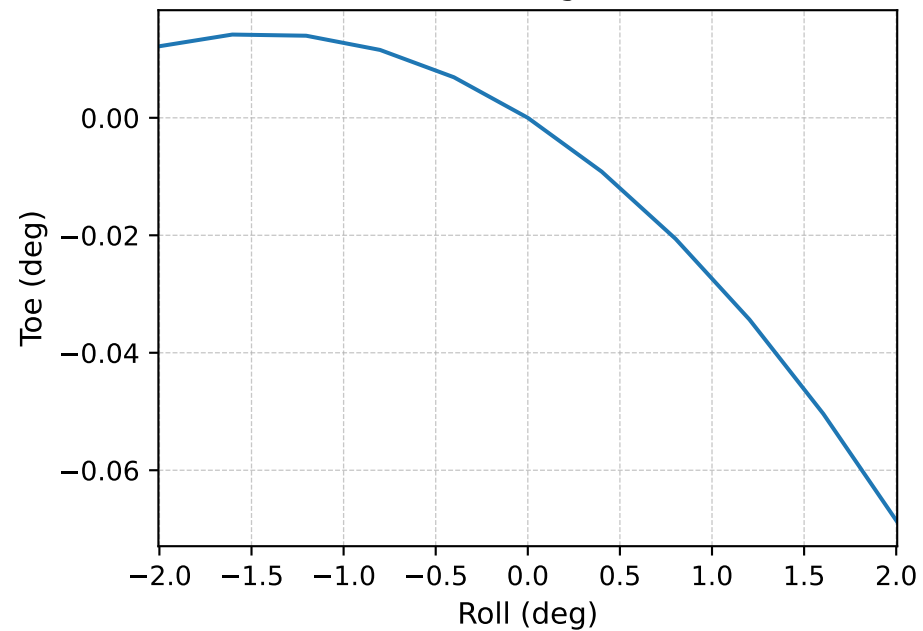
Front Right



Rear Left

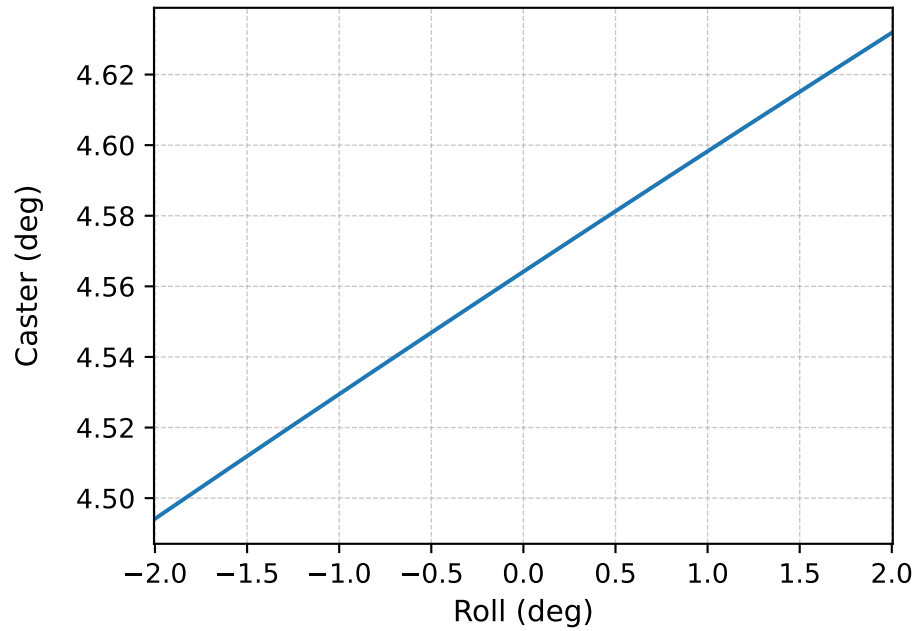


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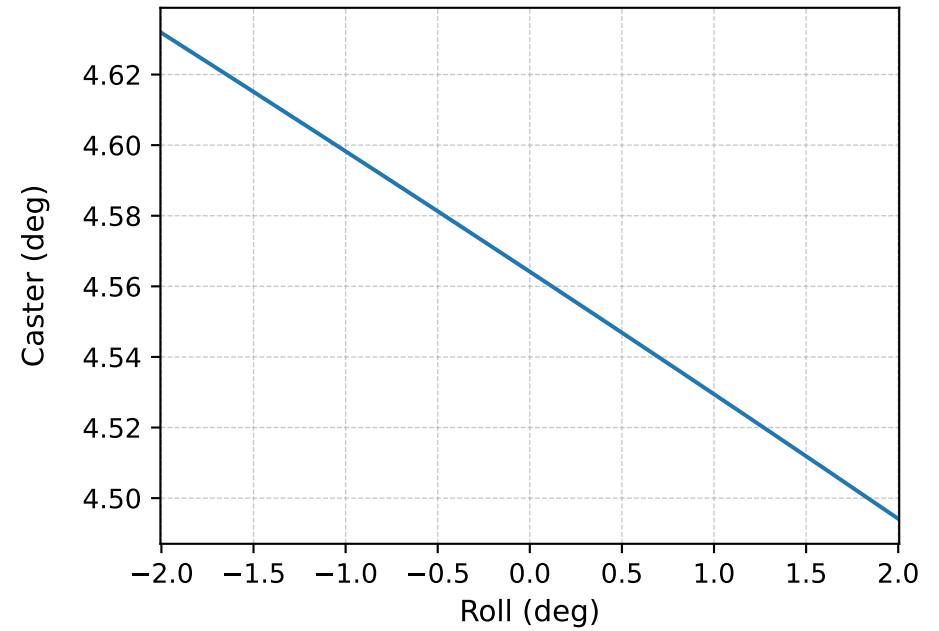


Caster vs Roll

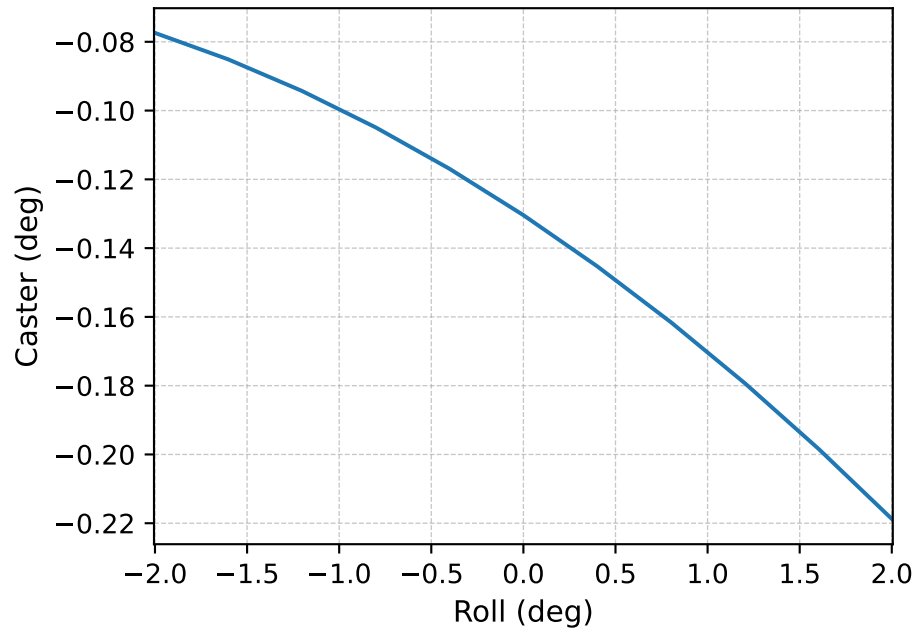
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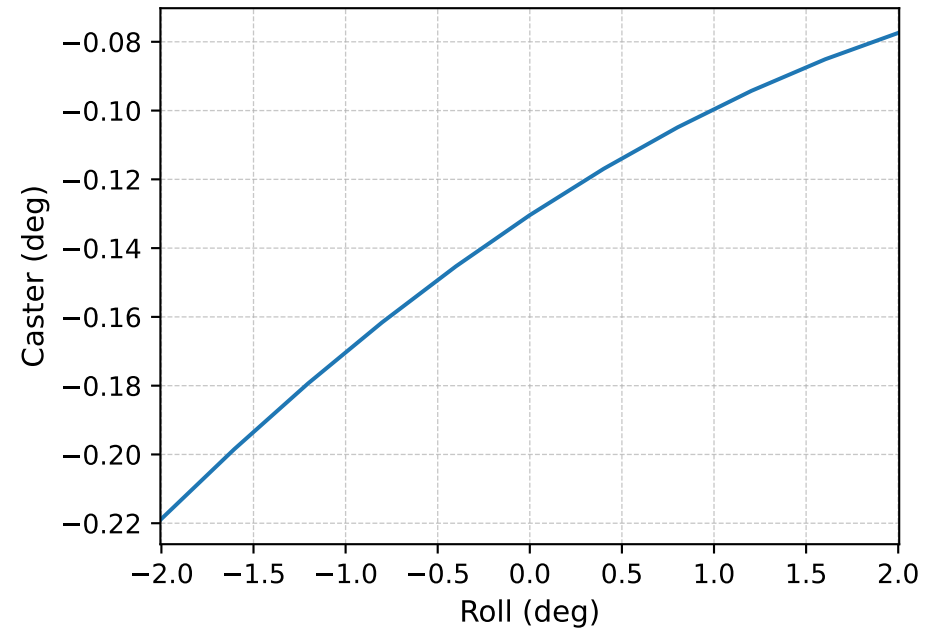
Front Right



Rear Left

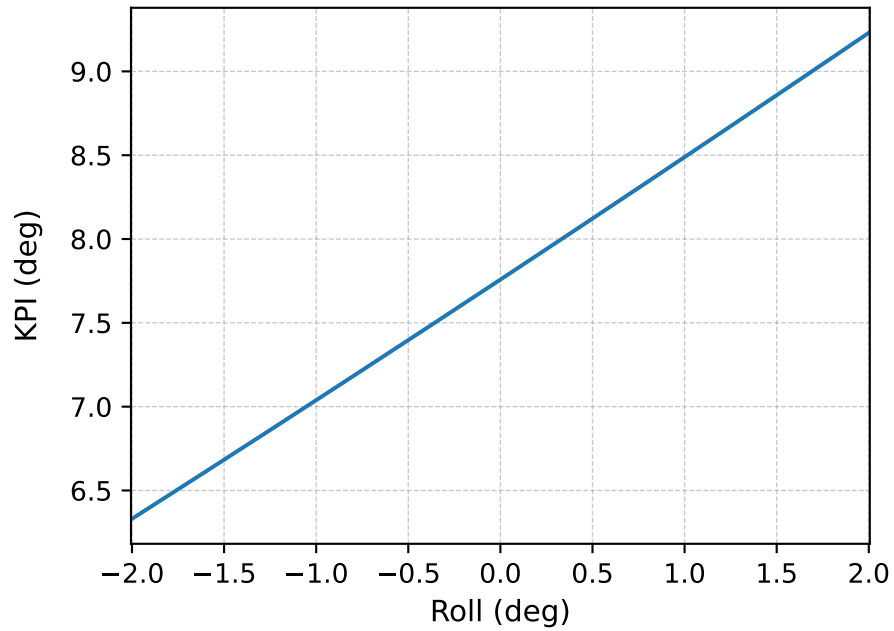


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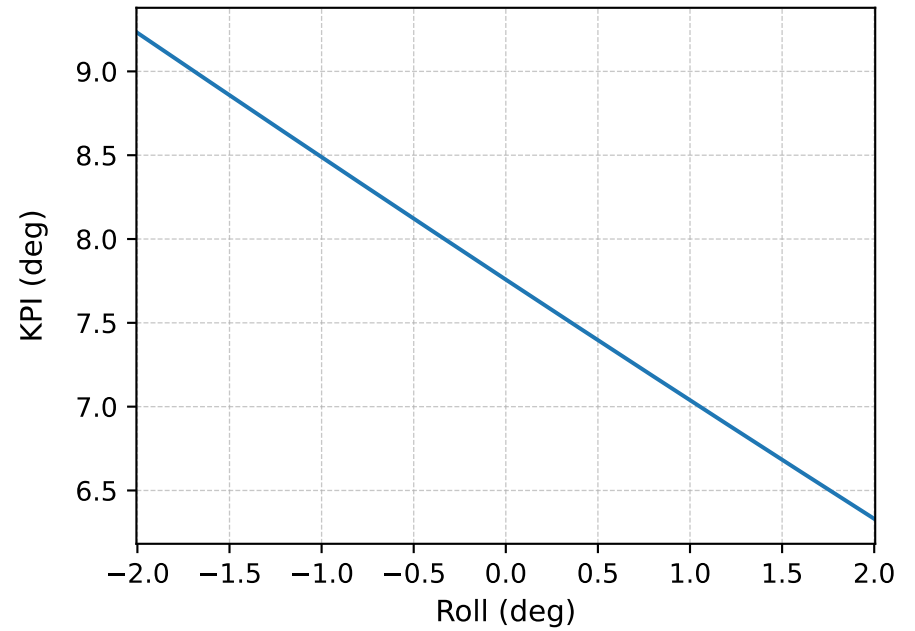


KPI vs Roll

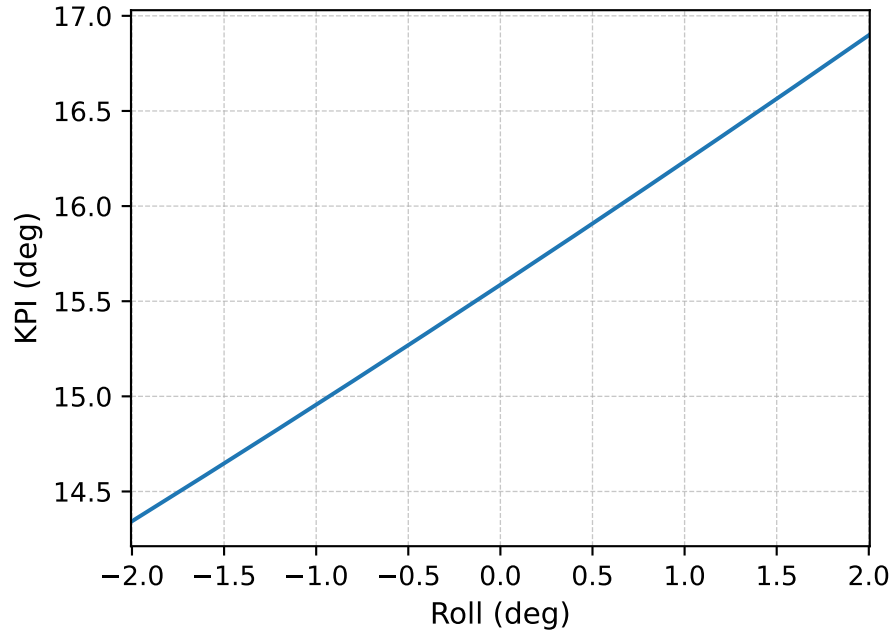
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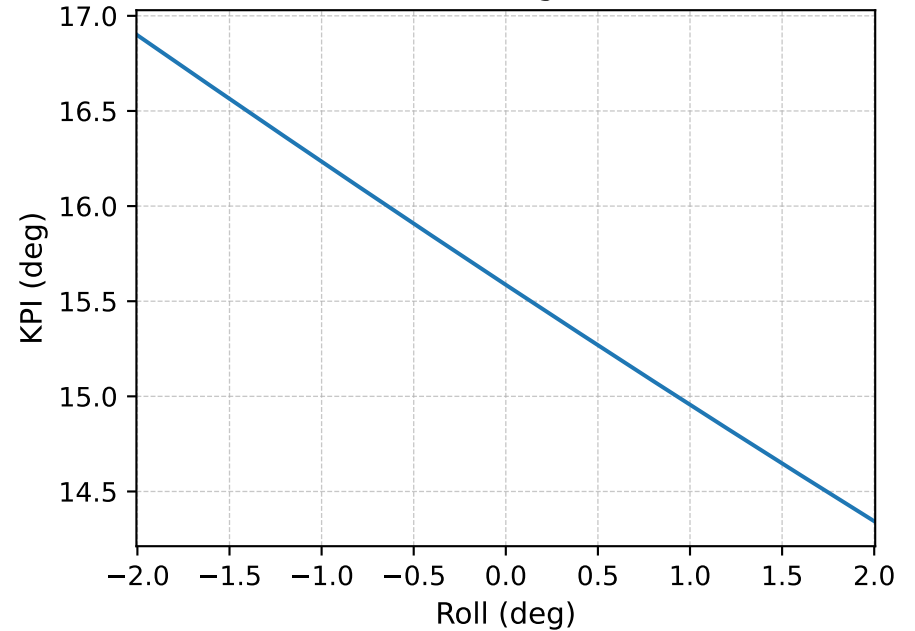
Front Right



Rear Left

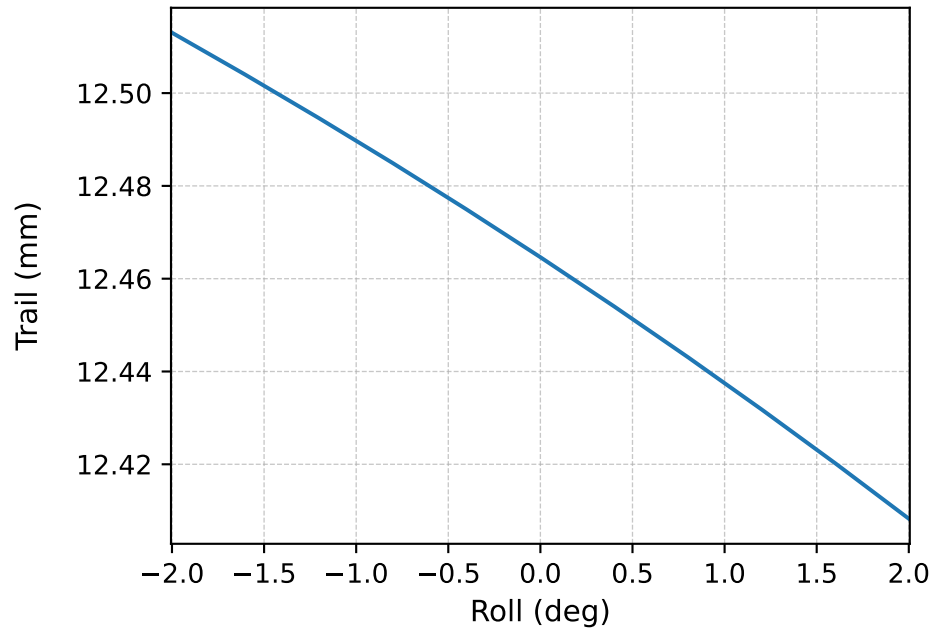


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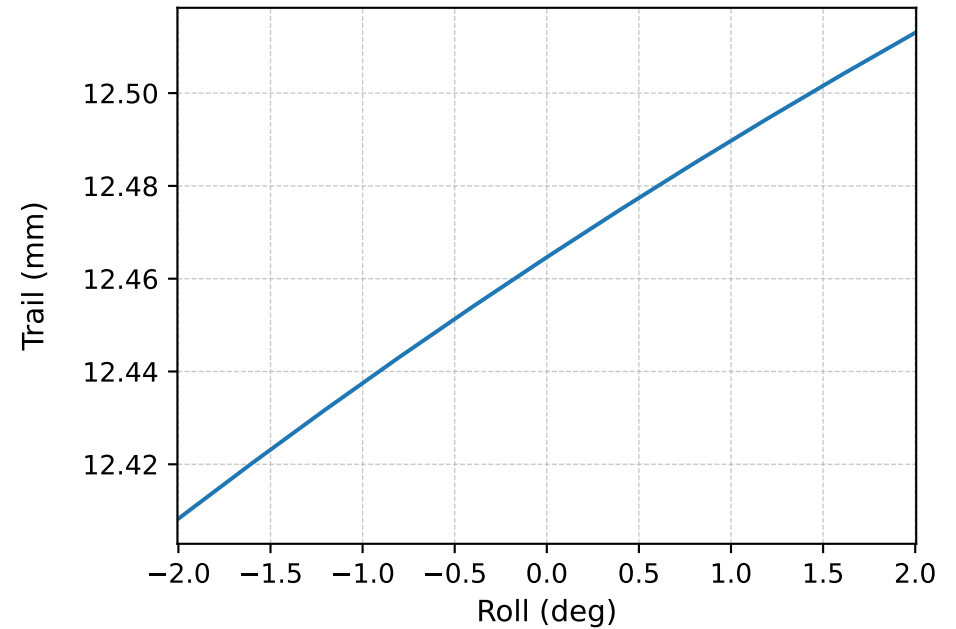


Mechanical Trail vs Roll

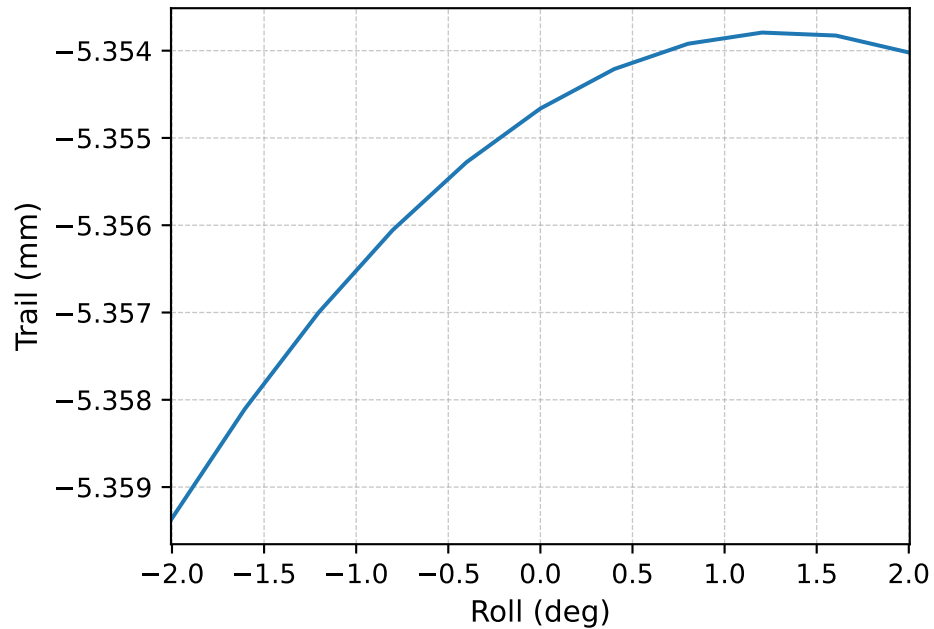
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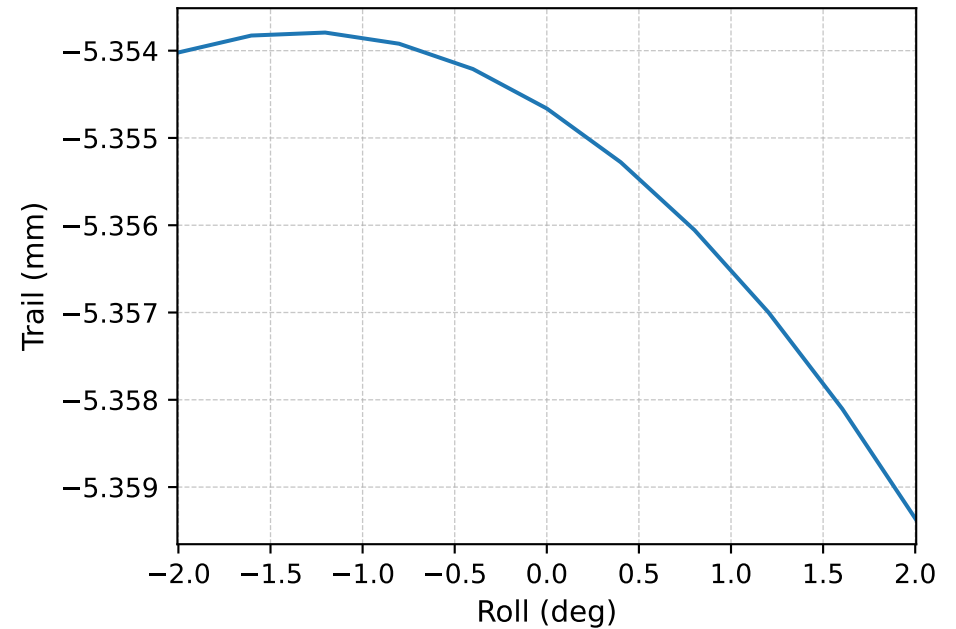
Front Right



Rear Left

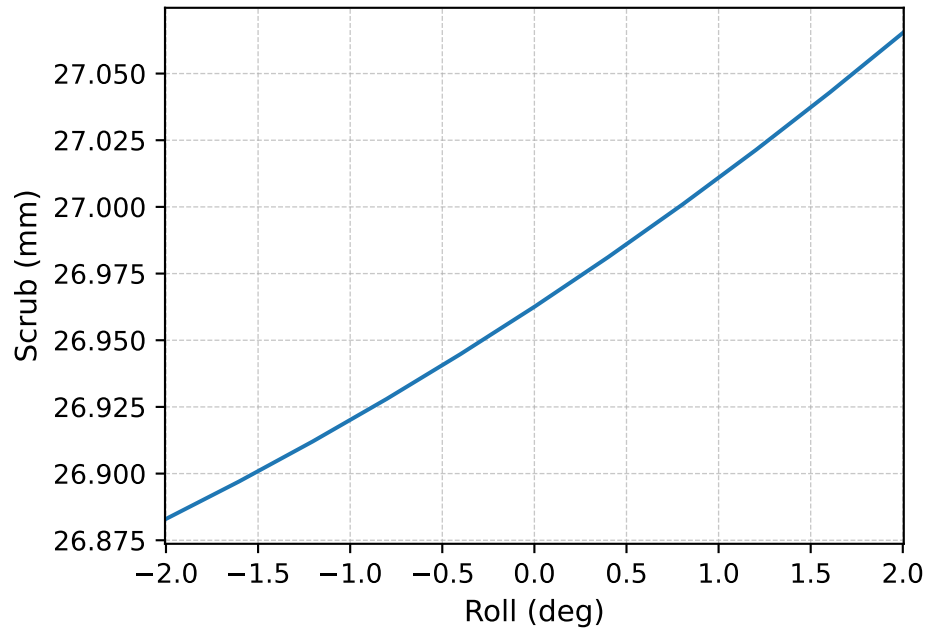


Rear Right

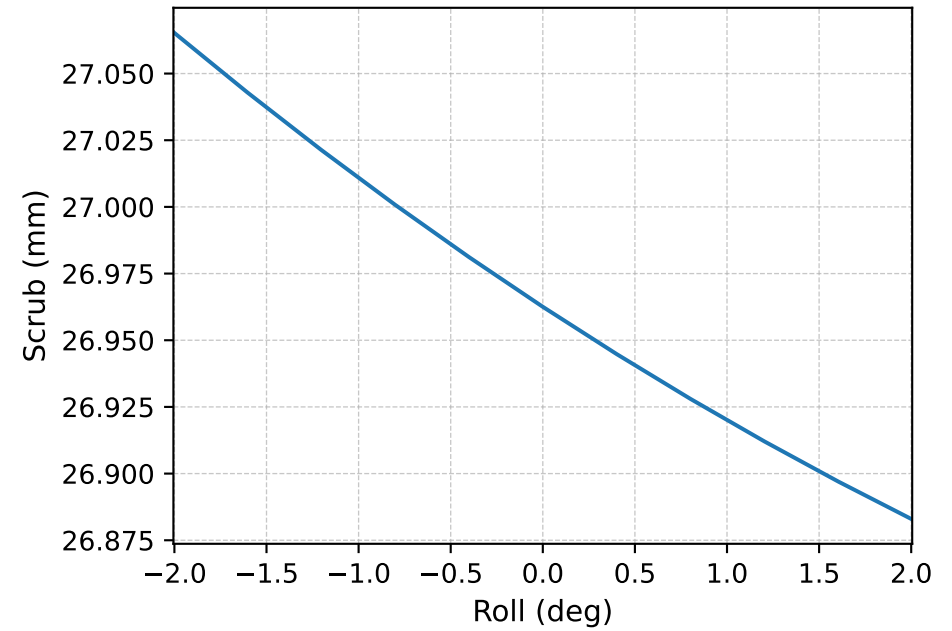


Mechanical Scrub vs Roll

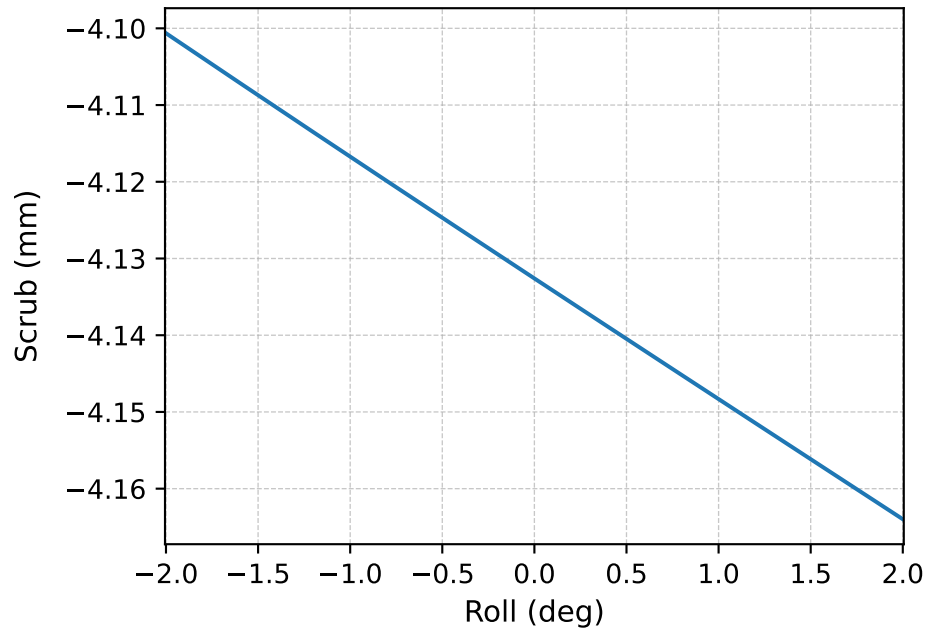
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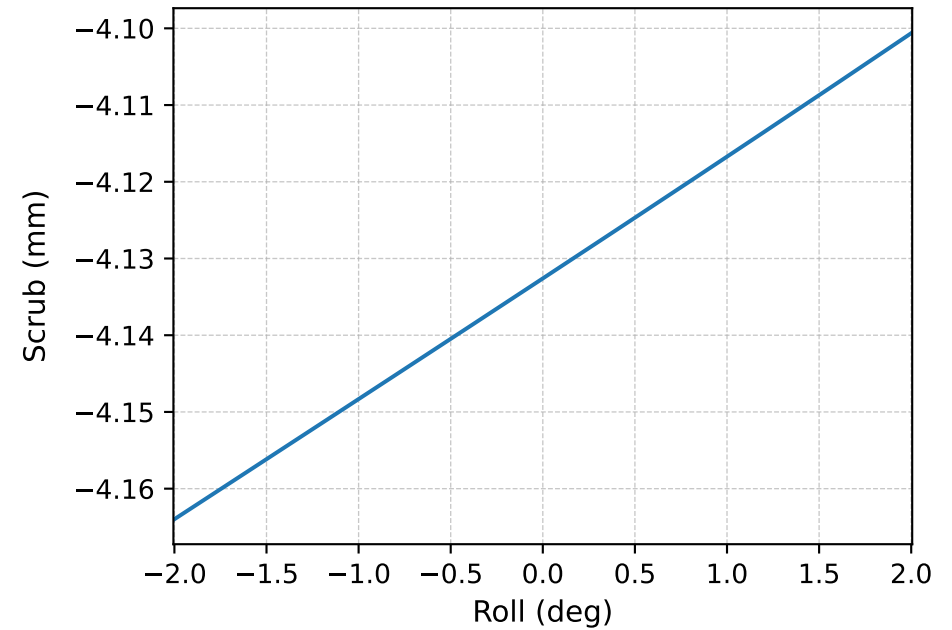
Front Right



Rear Left

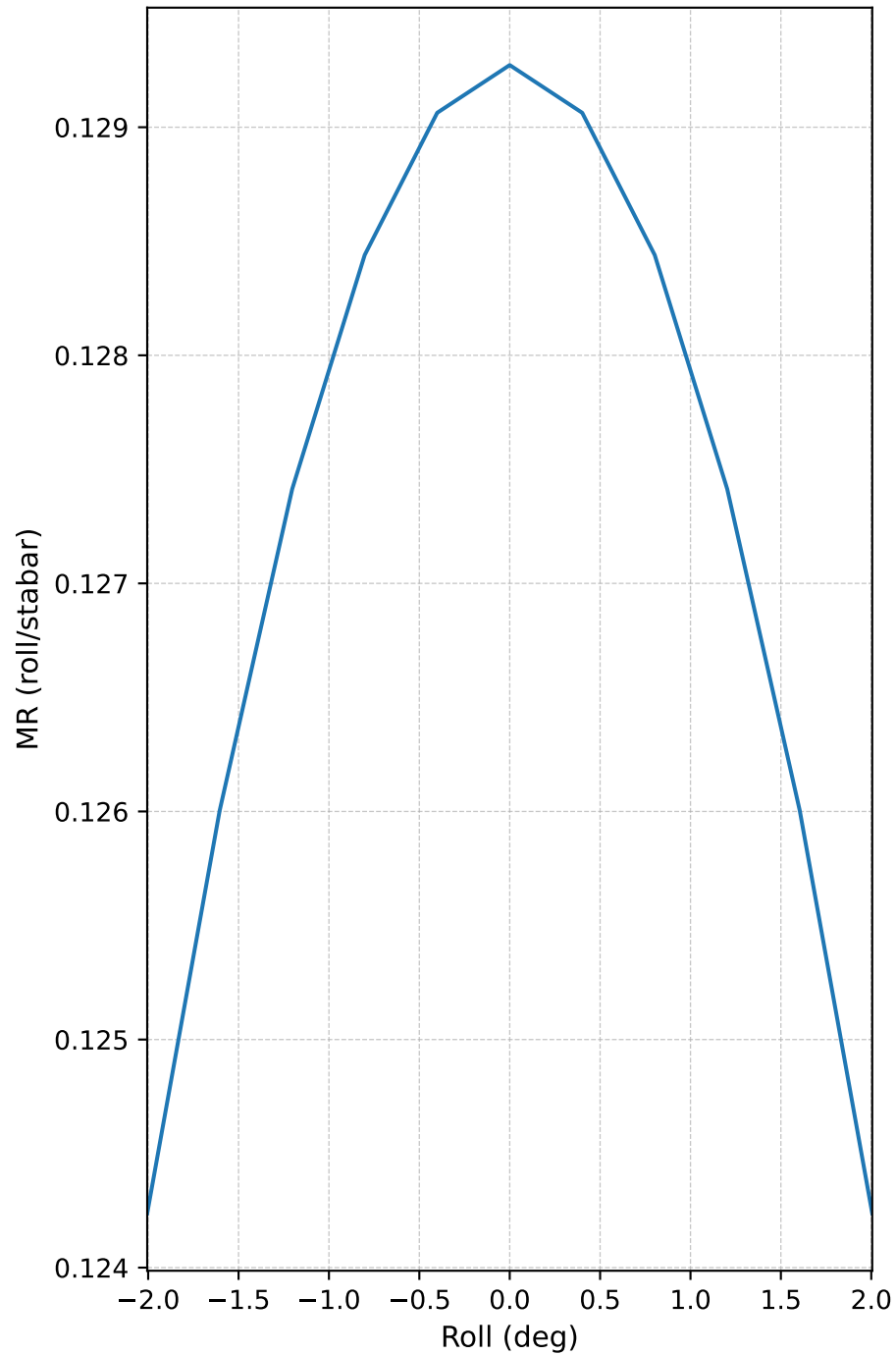


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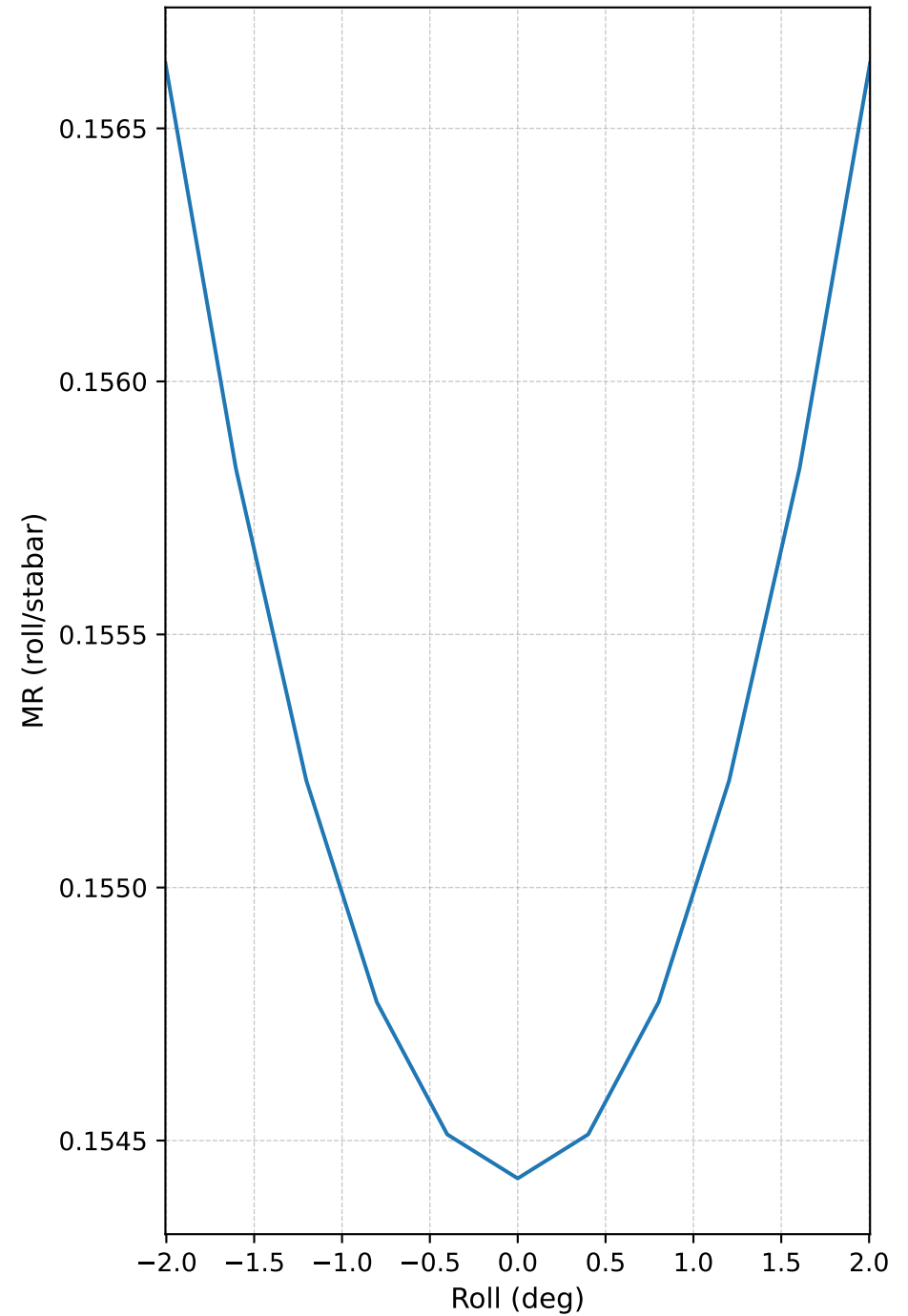


Stabar Motion Ratio vs Roll

Front

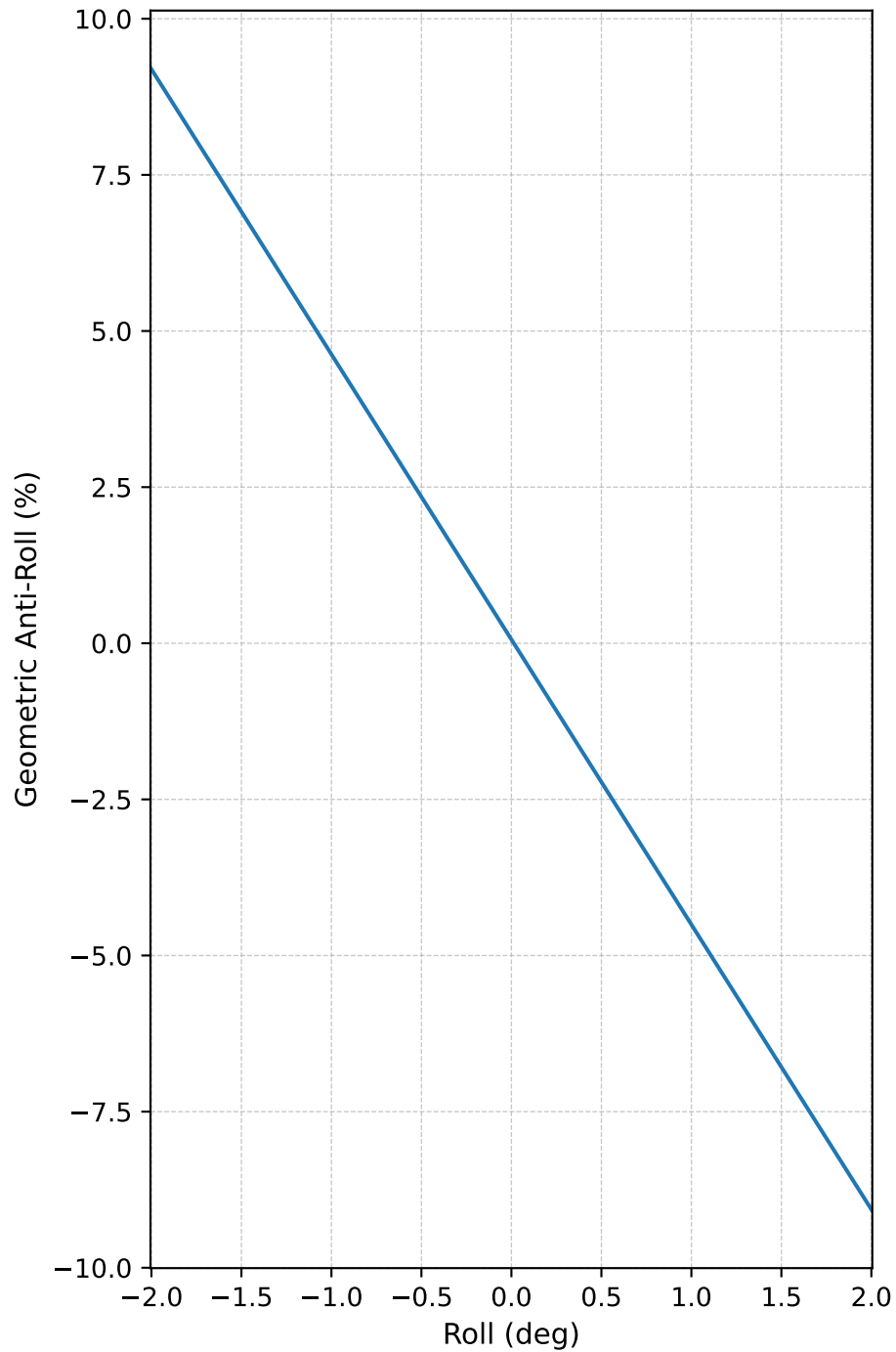


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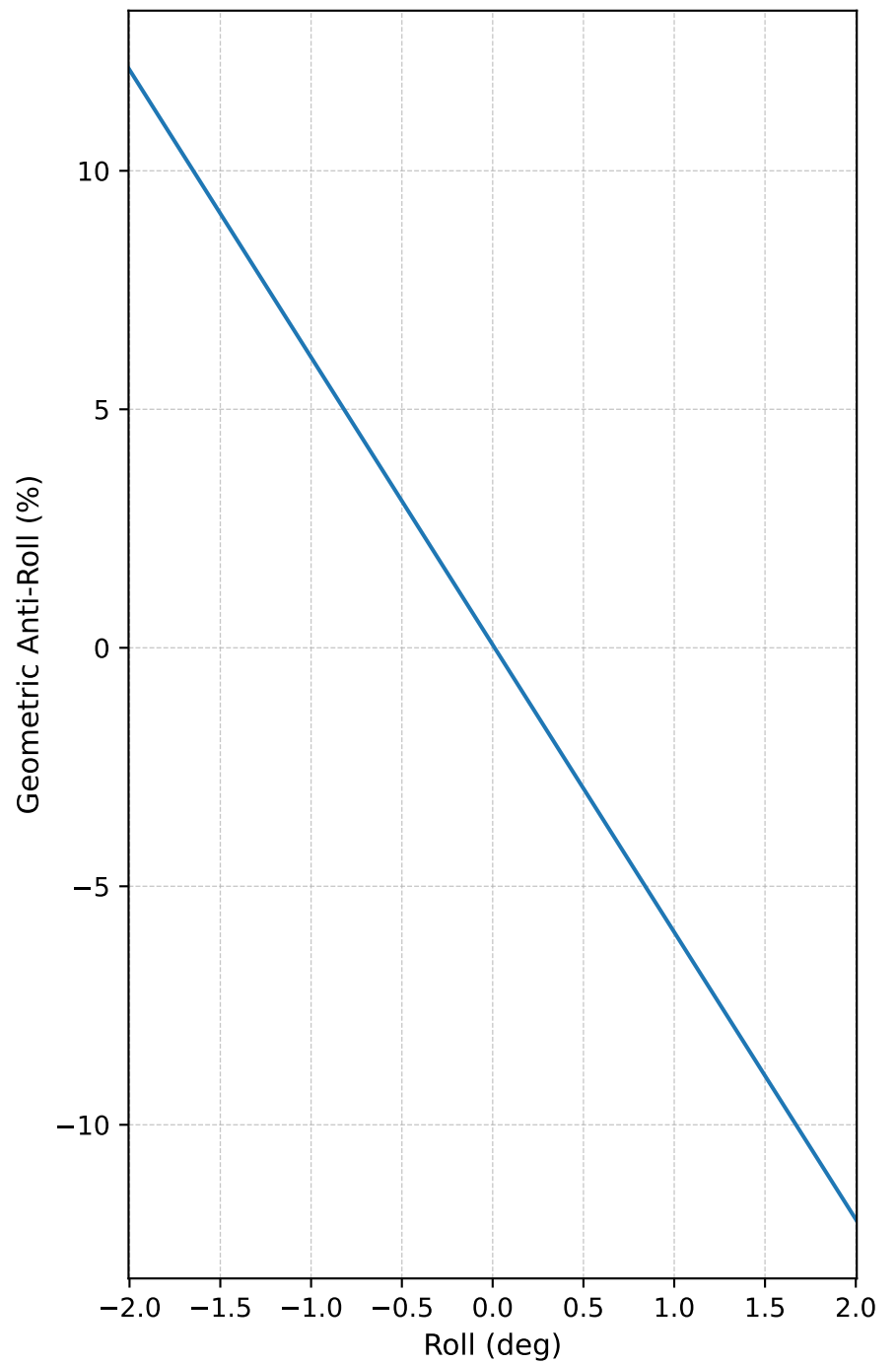


Geometric Anti-Roll vs Roll

Front



Rear



LLTD vs Roll

